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14 September 2026

Data 260 Homework 4

Personal Configuration and Domain

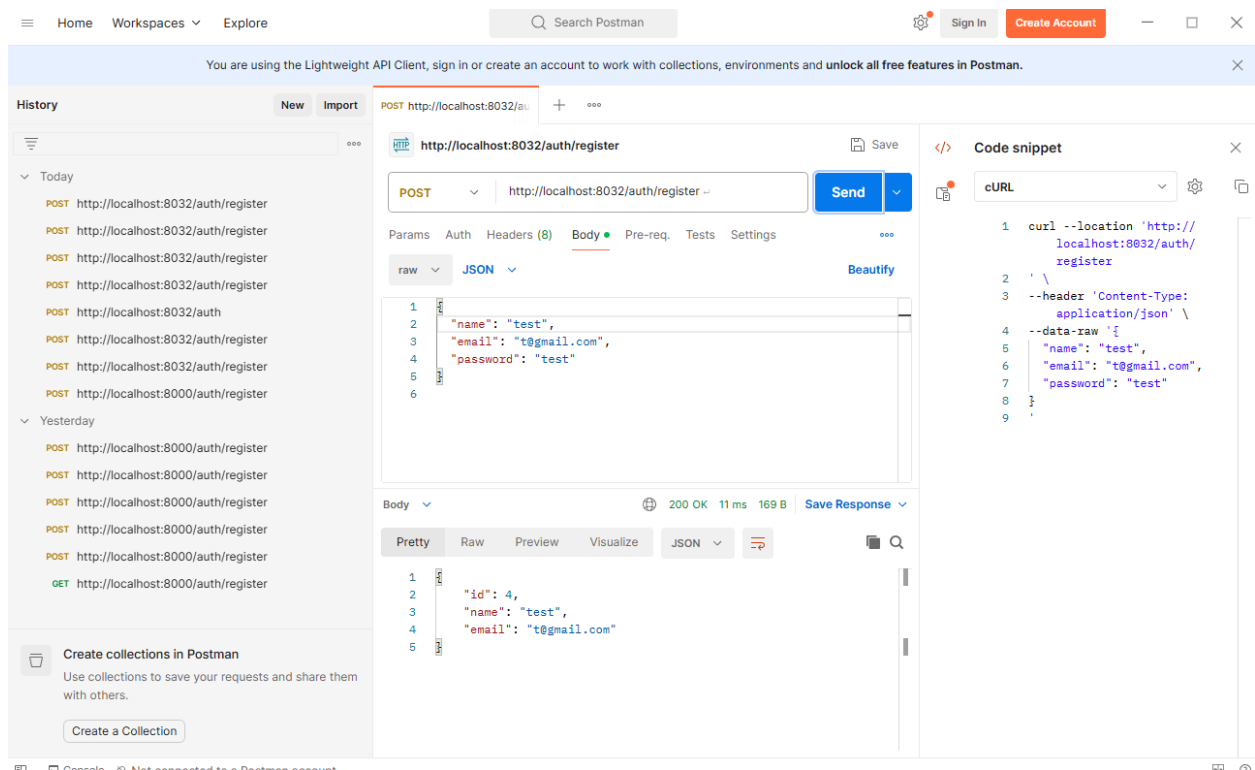
SID4	0932
PORT_BASE	8032
PREFIX	s0932
SEED	0932
VERIFY_SEED	260932
DOMAIN_ID	4 (open-source package vulnerabilities)

Hardware: nvidia geforce 4070, msi cpu

Local model used: ollama qwen3:8b

Tagged commit hash repo with most updated codes (submission on canvas only):

<https://github.com/n-k-leung/data260-0932>



Part 1 - React Client

This homework builds on HW3. Your existing FastAPI backend becomes the backend for the React client below, and your domain data moves to MySQL.

I used the same routes in homework 3 and fit it with the homework 4 demo. I used the homework 4 demo CSS style.

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04\frontend> npm run dev

> frontend@1.0.0 dev
> vite

VITE v5.4.21 ready in 210 ms

→ Local: http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04\backend> uvicorn app.main:app --reload --port 8032
INFO: Will watch for changes in these directories: ['C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04\backend']
INFO: Uvicorn running on http://127.0.0.1:8032 (Press CTRL+C to quit)
INFO: Started reloader process [38488] using StatReload
INFO: Started server process [36704]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:65512 - "GET /auth/me HTTP/1.1" 200 OK
INFO: 127.0.0.1:65513 - "GET /auth/me HTTP/1.1" 200 OK
INFO: 127.0.0.1:65512 - "GET /vuls HTTP/1.1" 200 OK
```

Add a Login page/component (Login.jsx) that allows a user to log in using email + Password.

```

data260-0932 > reports > hw04 > frontend > src > pages > Login.jsx > Login > handleSubmit
1  import React, { useState } from "react";
2  import { useNavigate } from "react-router-dom";
3  import { login } from "../api/vulApi.js";
4
5  export default function Login({ setAuth }) {
6    const navigate = useNavigate();
7    const [email, setEmail] = useState("");
8    const [password, setPassword] = useState("");
9    const [error, setError] = useState("");
10
11    async function handleSubmit(e) {
12      e.preventDefault();
13      setError("");
14      try {
15        const res = await login(email, password);
16        setAuth({ loggedIn: true, userId: res.user_id });
17        navigate("/");
18      } catch (err) {
19        setError("Invalid email or password.");
20      }
21    }
22
23    return (
24      <div className="card">
25        <div className="card-header">
26          <div className="page-title">Login</div>
27          <div className="subtitle">Log in to view and manage vulnerability records</div>
28        </div>
29
30        <div className="card-body">
31          <form className="form" onSubmit={handleSubmit}>
32            <label>
33              Email
34              <input
35                type="email"
36                value={email}
37                onChange={(e) => setEmail(e.target.value)} />
38            </label>
39
40            <label>
41              Password
42              <input
43                type="password"
44                value={password}
45                onChange={(e) => setPassword(e.target.value)} />
46            </label>
47
48            {error && <div className="notice">{error}</div>}
49
50            <button className="btn primary" type="submit">
51              Login
52            </button>
53          </form>
54        </div>
55      </div>
56    );
57  }
58
59

```

- After successful login, the backend will set an HTTP-only cookie containing a server-side session token.

Open Source Package Vulnerability Manager

[Home](#)[Add Vulnerability Record](#)[Logout](#)

Vulnerabilities

+ Add Record

Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session.

ID	PACKAGE NAME	SEVERITY	ACTIONS
5002	nodejs 2.11.1	High	UpdateDelete

InspectorConsoleDebuggerNetworkStyle EditorPerformanceMemoryStorageAccessibilityApplication

Cache Storage

Cookies

Indexed DB

Local Storage

Filter items

Name	Value	Domain	Path	Expires / Max-Age	Size	HttpOnly	Secure	SameSite	Last Accessed	Partition Key
session_id	b68196b8ad39470d514e3b35240fd2d27fa9e3b493193cf93203ca0...	localhost	/	Mon, 28 Sep 2026 16:46:09...	74	true	false	Lax	Mon, 28 Sep 2026 16:16:09...	

Open Source Package Vulnerability Manager

HomeAdd Vulnerability RecordLogout

Vulnerabilities

Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session.

+ Add Record

ID	PACKAGE NAME	SEVERITY	ACTIONS
5002	nodejs 2.11.1	High	UpdateDelete

InspectorConsoleDebuggerNetworkStyle EditorPerformanceMemoryStorageAccessibilityApplication

Filter URLs

Filter Headers

Filter Cookies

Filter Request

Filter Response

Filter Timings

Filter Stack Trace

Method	Domain	File	Initiator	Type	Transferred	Size	Headers	Cookies	Request	Response	Timings	Stack Trace
GET	localhost:5173	login	document	html	cached	585 B						
GET	localhost:5173	client	script	js	cached	137.73...						
GET	localhost:5173	main.jsx	script	js	cached	2.85 kB						
GET	localhost:5173	@react-refresh	script	js	cached	117.55...						
GET	localhost:5173	react_jsx-dev-runtime.js?v=0c4335a2	script	js	cached	0 B						
GET	localhost:5173	react.js?v=0c4335a2	script	js	cached	219 B						
GET	localhost:5173	react-dom_client.js?v=0c4335a2	script	js	cached	0 B						
GET	localhost:5173	react-router-dom.js?v=0c4335a2	script	js	cached	0 B						
GET	localhost:5173	app.jsx	script	js	cached	19.56 kB						
GET	localhost:5173	styles.css	script	js	cached	7.55 kB						
GET	localhost:5173	env.mjs	script	js	cached	4.02 kB						
GET	localhost:5173	chunk-DRWLMN53.js?v=0c4335a2	script	js	cached	0 B						
GET	localhost:5173	chunk-G3PMV6ZZ.js?v=0c4335a2	script	js	cached	0 B						
GET	localhost:5173	chunk-PJEEZAML.js?v=0c4335a2	script	js	cached	0 B						
GET	localhost:5173	Navbar.jsx	script	js	cached	10.09 kB						
GET	localhost:5173	Home.jsx	script	js	cached	18.71 kB						
GET	localhost:5173	CreateRecord.jsx	script	js	cached	10.04 kB						
GET	localhost:5173	UpdateRecord.jsx	script	js	cached	12.31 kB						
GET	localhost:5173	DeleteRecord.jsx	script	js	cached	9.88 kB						
GET	localhost:5173	Login.jsx	script	js	cached	11.94 kB						
GET	localhost:5173	vulApi.js	script	js	cached	5.83 kB						
GET	localhost:5173	/?token=70VzptVyUfft	client:536 (webso...	plain	163 B	0 B						
GET	localhost:5173	axios.js?v=0c4335a2	script	js	cached	0 B						
GET	localhost:8032	me	axios.js:1728 (xhr)	json	267 B	26 B						
GET	localhost:8032	me	axios.js:1728 (xhr)	json	267 B	26 B						
GET	localhost:5173	favicon.ico	img	plain	cached	0 B						
POST	localhost:8032	login	axios.js:1728 (xhr)	json	401 B	35 B						
GET	localhost:8032	vuls	axios.js:1728 (xhr)	json	232 B	2 B						
POST	localhost:8032	vuls	axios.js:1728 (xhr)	json	291 B	60 B						

Response Headers (366 B)

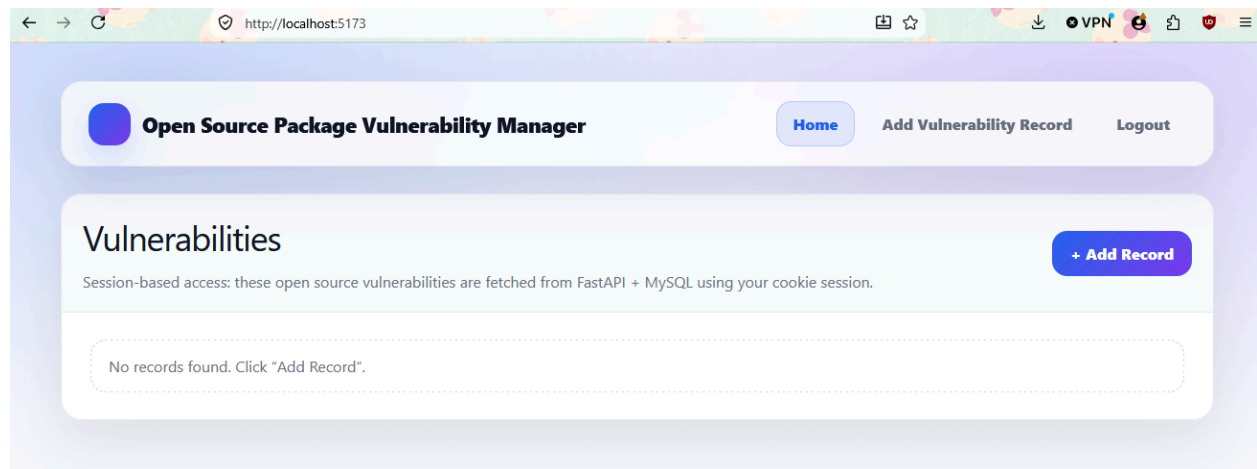
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:5173
content-length: 35
content-type: application/json
date: Mon, 28 Sep 2026 16:16:08 GMT
server: uvicorn
set-cookie: session_id=b68196b8ad39470d514e3b35240fd2d27a9e3b493193cf93203ca0149195827; HttpOnly; Max-Age=1800; Path=/; SameSite=lax
vary: Origin

Request Headers (502 B)

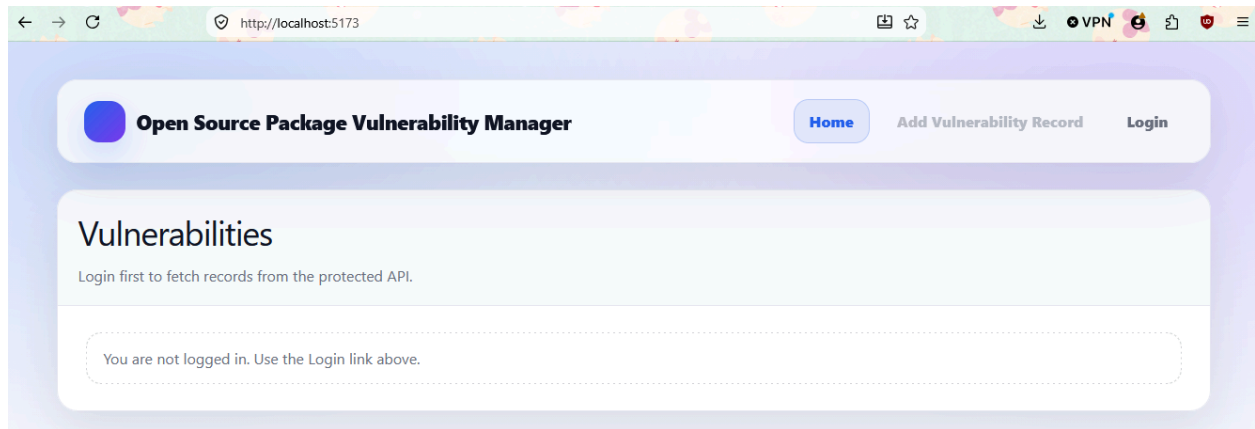
Accept: application/json, text/plain, */*
Accept-Encoding: gzip, deflate, br, zstd
Accept-Language: en-US,en;q=0.9
Connection: keep-alive
Content-Length: 41
Content-Type: application/json
Host: localhost:8032
Origin: http://localhost:5173/
Priority: u=0
Referer: http://localhost:5173/
Sec-Fetch-Dest: empty
Sec-Fetch-Mode: cors
Sec-Fetch-Site: same-site
Sec-GPC: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0

Status	Method	Domain	File	Initiator	Type	Transferred	Size
304	GET	localhost:5173	/	document	html	cached	585 B
304	GET	localhost:5173	client	script	js	cached	137.73...
304	GET	localhost:5173	main.jsx	script	js	cached	2.85 kB
304	GET	localhost:5173	@react-refresh	script	js	cached	117.55...
200	GET	localhost:5173	react_jsx-dev-runtime.js?v=0c4335a2	script	js	cached	0 B
200	GET	localhost:5173	react.js?v=0c4335a2	script	js	cached	219 B
200	GET	localhost:5173	react-dom_client.js?v=0c4335a2	script	js	cached	0 B
200	GET	localhost:5173	react-router-dom.js?v=0c4335a2	script	js	cached	0 B
304	GET	localhost:5173	app.jsx	script	js	cached	19.56 kB
304	GET	localhost:5173	styles.css	script	js	cached	7.55 kB
304	GET	localhost:5173	env.mjs	script	js	cached	4.02 kB
200	GET	localhost:5173	chunk-DRWLMN53.js?v=0c4335a2	script	js	cached	0 B
200	GET	localhost:5173	chunk-G3PMV62Z.js?v=0c4335a2	script	js	cached	0 B
200	GET	localhost:5173	chunk-PJEEZAML.js?v=0c4335a2	script	js	cached	0 B
304	GET	localhost:5173	Navbar.jsx	script	js	cached	10.09 kB
304	GET	localhost:5173	Home.jsx	script	js	cached	18.71 kB
304	GET	localhost:5173	CreateRecord.jsx	script	js	cached	10.04 kB
304	GET	localhost:5173	UpdateRecord.jsx	script	js	cached	12.31 kB
304	GET	localhost:5173	DeleteRecord.jsx	script	js	cached	9.68 kB
304	GET	localhost:5173	Login.jsx	script	js	cached	11.94 kB
304	GET	localhost:5173	vulApi.js	script	js	cached	5.83 kB
401	GET	localhost:5173	/?token=7OVzptVyUfft	client:536 (webso...	plain	163 B	0 B
200	GET	localhost:5173	axios.js?v=0c4335a2	script	js	cached	0 B
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B

- Only logged-in users should be allowed to view the record list and perform CRUD Operations.



- If the user is not logged in: show a message "Login required", and/or disable/hide navigation buttons like "Add Record".



I. Home Page

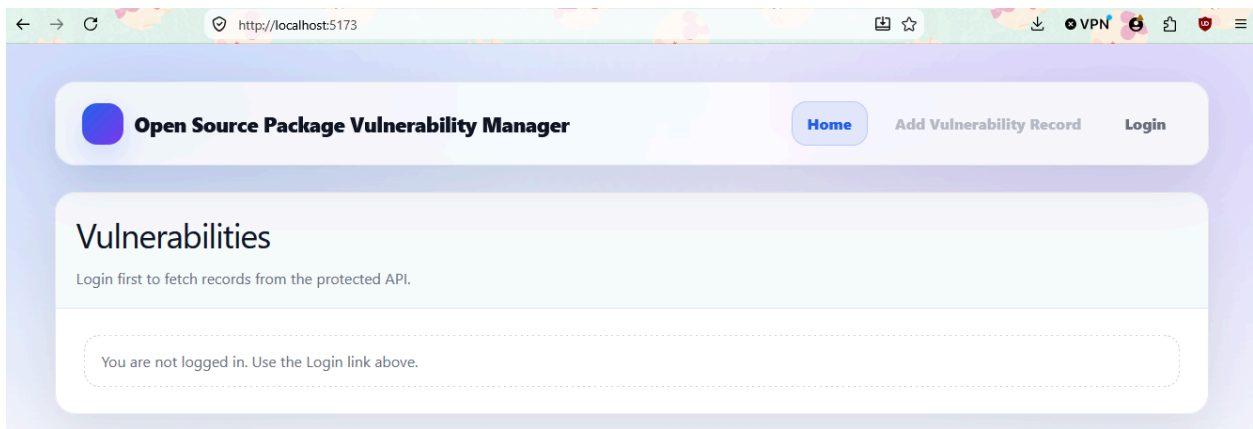
Write the necessary code to create the Home.jsx component which will display all the records. The Home component should be rendered when the user is on the root route /.

```
data260-0932 > reports > hw04 > frontend > src > pages > Home.jsx > Home
1  import React from "react";
2  import { Link } from "react-router-dom";
3
4  export default function Home({ records, loading, auth }) {
5    // If not logged in, show a clear message (since backend is protected)
6    if (!auth.loggedIn) {
7      return (
8        <div className="card">
9          <div className="card-header">
10             <div>
11               <div className="page-title">Vulnerabilities</div>
12               <div className="subtitle">
13                 Login first to fetch records from the protected API.
14               </div>
15             </div>
16           </div>
17           <div className="card-body">
18             <div className="notice">
19               You are not logged in. Use the Login link above.
20             </div>
21           </div>
22         </div>
23       );
24     }
25
26     // logged in: show vulnerabilities table
27
28     return (
29       <div className="card">
30         <div className="card-header">
31           <div>
32             <div className="page-title">Vulnerabilities</div>
33             <div className="subtitle">
34               Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session.
35             </div>
36           </div>
37           <Link className="btn primary" to="/create">
38             + Add Record
39           </Link>
40         </div>
41         <div className="card-body">
42           <div>
43             <div>
44               <div>
45                 <div className="notice">loading records....</div>
46               </div>
47             </div>
48             <div>
49               <div>
50                 <div className="notice">No records found. Click "Add Record".</div>
51               </div>
52             </div>
53             <div>
54               <div>
55                 <div>
56                   <div className="table-wrap">
57                     <table className="table">
58                       <thead>
59                         <tr>
60                           <th>ID</th>
61                           <th>Package Name</th>
62                           <th>Severity</th>
63                           <th>Actions</th>
64                         </tr>
65                       </thead>
66                       <tbody>
```

```

61     {records.map((u) => (
62         <tr key={u.id}>
63             <td>{u.id}</td>
64             <td>{u.package_name}</td>
65             <td>{u.severity}</td>
66             <td className="actions">
67                 <Link className="btn" to={` /update/${u.id}`}>
68                     Update
69                 </Link>
70                 <Link className="btn danger" to={` /delete/${u.id}`}>
71                     Delete
72                 </Link>
73             </td>
74         </tr>
75     )]}
76 </tbody>
77 </table>
78 </div>
79 )}
80 </div>
81 </div>
82 );
83

```



II. Add a New Record

Write the necessary code to create the `CreateRecord.jsx` component. The component should be rendered when the user is on the `/create` route.

```

data260-0932 > reports > hw04 > frontend > src > pages > CreateRecord.jsx > CreateRecord
1  import React, { useState } from "react";
2
3  export default function CreateRecord({ onAdd }) {
4    const [package_name, setPackageName] = useState("");
5    const [severity, setSeverity] = useState("");
6
7    async function handleSubmit(e) {
8      e.preventDefault();
9      await onAdd({ package_name: package_name, severity:severity });
10   }
11
12   return (
13     <div className="card">
14       <div className="card-header">
15         <div className="page-title">Add Record</div>
16         <div className="subtitle">Enter record details below</div>
17       </div>
18
19       <div className="card-body">
20         <form className="form" onSubmit={handleSubmit}>
21           <label>
22             Package Name
23             <input
24               value={package_name}
25               onChange={(e) => setPackageName(e.target.value)}
26             />
27           </label>
28
29           <label>
30             Severity
31             <input
32               value={severity}
33               onChange={(e) => setSeverity(e.target.value)}
34             />
35           </label>
36
37           <button className="btn primary" type="submit">
38             Add Record
39           </button>
40         </form>
41       </div>
42     </div>
43   );
44 }

```

The component should accept props to add the new record and have the following:

- An input field for entering the entity's primary field.
- An input field for entering the entity's secondary field.
- A submit button labeled appropriately for your domain.

The screenshot shows a web browser window with the address bar displaying `http://localhost:5173/create`. The page title is "Open Source Package Vulnerability Manager". The navigation bar includes links for "Home", "Add Vulnerability Record", and "Logout". The main content area is titled "Add Record" with a subtitle "Enter record details below". The form contains two input fields: "Package Name" with the value "nodejs 22.1" and "Severity" with the value "High". A blue "Add Record" button is positioned at the bottom of the form.

Open Source Package Vulnerability Manager

Home Add Vulnerability Record Logout

Add Record

Enter record details below

Package Name

nodejs 22.1

Severity

High

Add Record

- When the user clicks the submit button:
- A new record should be added to the list with an auto-incremented ID.
- The user should be redirected to the home page to see the updated list.

The screenshot shows a web application titled "Open Source Package Vulnerability Manager" running on `http://localhost:5173`. The application has a navigation bar with links for "Home", "Add Vulnerability Record", and "Logout". The main section is titled "Vulnerabilities" and includes a sub-header: "Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session." Below this is a table of vulnerabilities.

ID	PACKAGE NAME	SEVERITY	ACTIONS
1	nodejs 22.1	High	Update Delete

Below the table is a network inspector showing a list of requests. The selected request is a GET request to `localhost:5173/react-router-dom.js?v=0c4335a2` with a status of 200. The response is a JSON object:

```

{
  "id": 1,
  "package_name": "nodejs 22.1",
  "severity": "High"
}

```

III. Update Record

Write the necessary code to create the `UpdateRecord.jsx` component. The component should be rendered when the user is on the `/update` route.

```

data260-0932 > reports > hw04 > frontend > src > pages > UpdateRecord.jsx > UpdateRecord
1  import React, { useEffect, useState } from "react";
2  import { useParams } from "react-router-dom";
3  import { fetchVulById } from "../api/vulApi.js";
4
5  export default function UpdateRecord({ onUpdate }) {
6    const { id } = useParams();
7    const recordId = Number(id);
8
9    const [package_name, setPackageName] = useState("");
10   const [severity, setSeverity] = useState("");
11   const [loading, setLoading] = useState(true);
12
13   useEffect(() => {
14     (async () => {
15       try {
16         setLoading(true);
17         const record = await fetchVulById(recordId);
18         setPackageName(record.package_name);
19         setSeverity(record.severity);
20       } catch (e) {
21         console.error(e);
22       } finally {
23         setLoading(false);
24       }
25     })();
26   }, [recordId]);
27
28   async function handleSubmit(e) {
29     e.preventDefault();
30     await onUpdate(recordId, { package_name, severity });
31   }
32
33   if (loading) return <p>Loading record...</p>;
34
35   return (
36     <div className="card">
37       <div className="card-header">
38         <div className="page-title">Update Record (ID: {recordId})</div>
39       </div>
40
41       <div className="card-body">
42         <form className="form" onSubmit={handleSubmit}>
43           <label>
44             Package Name
45             <input
46               value={package_name}
47               onChange={(e) => setPackageName(e.target.value)}
48             />
49           </label>
50
51           <label>
52             Severity
53             <input
54               value={severity}
55               onChange={(e) => setSeverity(e.target.value)}
56             />
57           </label>
58
59           <button className="btn primary" type="submit">
60             Update Record
61           </button>
62         </form>
63       </div>
64     </div>
65   );
66

```

The component should accept props to update the record and have the following:

- An input field for the primary field.
- An input field for the secondary field.
- A submit button labeled appropriately for your domain.

A screenshot of a web browser showing the 'Update Record (ID: 1)' page of the 'Open Source Package Vulnerability Manager'. The browser's address bar shows 'http://localhost:5173/update/1'. The page has a light purple header with the application name and navigation links: 'Home', 'Add Vulnerability Record', and 'Logout'. The main content area has a title 'Update Record (ID: 1)'. Below the title, there are two input fields: 'Package Name' with the value 'nodejs 22.1' and 'Severity' with the value 'High'. At the bottom of the form is a blue button labeled 'Update Record'.

When the user clicks the submit button, the record should be updated through the backend API and persisted in MySQL and the user should be redirected to the home page showing the updated list.

A second screenshot of the same 'Update Record (ID: 1)' page. The 'Package Name' input field now contains 'nodejs 22.3', while the 'Severity' field remains 'High'. The blue 'Update Record' button is still at the bottom. The browser address bar and header are identical to the first screenshot.

The screenshot shows a web application titled "Open Source Package Vulnerability Manager" with a navigation bar containing "Home", "Add Vulnerability Record", and "Logout". The main section is titled "Vulnerabilities" and includes a sub-header: "Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session." A "+ Add Record" button is present. Below this is a table with the following data:

ID	PACKAGE NAME	SEVERITY	ACTIONS
1	nodejs 22.3	High	Update Delete

The bottom part of the image shows the browser's developer tools network tab. It displays a list of network requests. The selected request is a GET request to "http://localhost:5173/react-router-dom.js?v=0c4335a2". The response is a JSON object with the following properties:

```

{
  "id": 1,
  "package_name": "nodejs 22.3",
  "severity": "High"
}

```

IV. Delete the Record

Write the necessary code to create the DeleteRecord.jsx component. The component should be rendered when the user is on the /delete route.

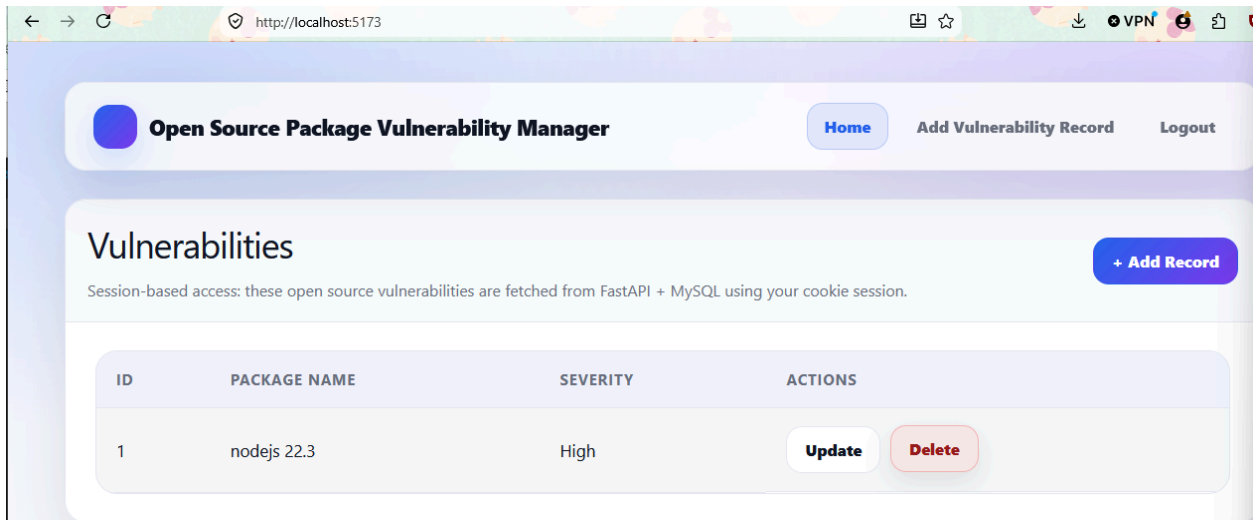
```

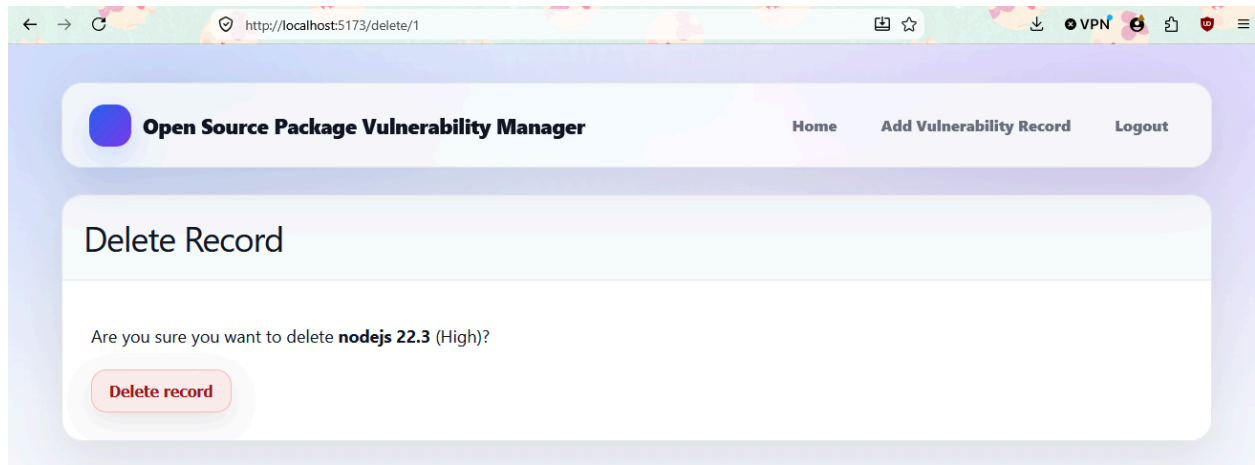
data260-0932 > reports > hw04 > frontend > src > pages > DeleteRecord.jsx > DeleteRecord > useEffect() callback >
1 import React, { useEffect, useState } from "react";
2 import { useParams } from "react-router-dom";
3 import { fetchVulById } from "../api/vulApi.js";
4
5 export default function DeleteRecord({ onDelete }) {
6   const { id } = useParams();
7   const recordId = Number(id);
8
9   const [record, setRecord] = useState(null);
10
11   useEffect(() => {
12     (async () => {
13       try {
14         const data = await fetchVulById(recordId);
15         setRecord(data);
16       } catch {
17         setRecord(null);
18       }
19     })();
20   }, [recordId]);
21
22   async function handleDelete() {
23     await onDelete(recordId);
24   }
25
26   return (
27     <div className="card">
28       <div className="card-header">
29         <div className="page-title">Delete Record</div>
30       </div>
31
32       <div className="card-body">
33         {record ? (
34           <>
35             <p style={{ fontSize: "18px", marginBottom: "24px" }}>
36               Are you sure you want to delete <strong>{record.package_name}</strong>
37             </p>
38
39             <button className="btn danger" onClick={handleDelete}>
40               Delete record
41             </button>
42           </>
43         ) : (
44           <div className="notice">
45             record not found (or already deleted).
46           </div>
47         )}
48       </div>
49     </div>
50   );
51 }

```

The component should accept props to delete a record and have the following:

- A button labeled appropriately for your domain (e.g., "Delete").





When the user clicks the delete button, the record should be deleted through the backend API and removed from MySQL and the user should be redirected to the home page showing the updated list.

The screenshot shows a web browser at `http://localhost:5173` displaying the 'Open Source Package Vulnerability Manager' application. The application has a navigation bar with 'Home', 'Add Vulnerability Record', and 'Logout' buttons. The main section is titled 'Vulnerabilities' and contains a message: 'Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session.' Below this, a dashed box indicates 'No records found. Click "Add Record".' A '+ Add Record' button is visible in the top right of the main section.

The browser's developer tools are open to the 'Network' tab, showing a list of requests. The selected request is a GET request to `localhost:5173/vulApi.js` with a status of 200. The response is a JSON object with the following properties:

```

{
  "id": 1,
  "package_name": "nodejs 22.3",
  "severity": "High"
}

```

The 'Network' tab also shows a list of other requests, including GET requests to `localhost:5173` and `localhost:8032`, and a DELETE request to `localhost:8032`.

V. For all the above components, make sure to:

- Pass props for the Create, Update, and Delete components.

```

72 // Create (props)
73 async function onAdd(newRecord) {
74   const created = await createVul(newRecord);
75   setRecords(prev => [...prev, created]);
76   navigate("/");
77 }
78
79 // Update (props)
80 async function onUpdate(id, updatedRecord) {
81   const updated = await updateVul(id, updatedRecord);
82   setRecords(prev => prev.map(u => (u.id === id ? updated : u)));
83   navigate("/");
84 }
85
86 // Delete (props)
87 async function onDelete(id) {
88   await deleteVul(id);
89   setRecords(prev => prev.filter(u => u.id !== id));
90   navigate("/");
91 }
92
93 return (
94   <div className="container">
95     <Navbar auth={auth} setAuth={setAuth}/>
96
97     /* Login session demo UI */
98     /* <Navbar auth={auth} setAuth={setAuth} /> */
99
100     <Routes>
101       <Route path="/" element=<Home records={records} loading={loading} auth={auth} /> />
102       <Route path="/login" element=<Login setAuth={setAuth}/> />
103       <Route path="/create" element=<RequireAuth auth={auth}><CreateRecord onAdd={onAdd} auth={auth} /></RequireAuth> />
104       <Route path="/update/:id" element=<RequireAuth auth={auth}><UpdateRecord onUpdate={onUpdate} auth={auth} /></RequireAuth> />
105       <Route path="/delete/:id" element=<RequireAuth auth={auth}><DeleteRecord onDelete={onDelete} auth={auth} /></RequireAuth> />
106     </Routes>
107   </div>
108 );
109

```

- Use hooks like useState and useEffect wherever necessary.

```

export default function App() {
  const navigate = useNavigate();
  // Auth state (cookie session is checked inside LoginBar via /auth/me)
  const [auth, setAuth] = useState({ loggedIn: false, userId: null });

  // Records data
  const [records, setRecords] = useState([]);
  const [loading, setLoading] = useState(false);

  useEffect(() => {
    (async () => {
      try {
        const data = await me();
        setAuth({ loggedIn: true, userId: data.user_id });
      } catch {
        setAuth({ loggedIn: false, userId: null });
      }
    })();
  }, []);

  // Fetch records only when logged in
  useEffect(() => {
    (async () => {
      if (!auth.loggedIn) {
        setRecords([]);
        setLoading(false);
        return;
      }
      try {
        setLoading(true);
        const data = await fetchVuls();
        setRecords(data);
      } catch (e) {
        console.error("fetchVuls failed:", e);
        // If session expired, backend returns 401; UI will show not logged in after next /auth/me check.
      } finally {
        setLoading(false);
      }
    })();
  }, [auth.loggedIn]);
}

```

- Use React Router (react-router-dom) for routing.

data260-0932 > reports > hw04 > frontend > src > app.jsx > App > useEffect() callback > <function>

```

1 import React, { useEffect, useState } from "react";
2 import { Routes, Route, useNavigate } from "react-router-dom";
3

```

data260-0932 > reports > hw04 > frontend > src > main.jsx

```

1 import React from "react";
2 import ReactDOM from "react-dom/client";
3 import { BrowserRouter } from "react-router-dom";

```

```
data260-0932 > reports > hw04 > frontend > src > components >  Navbar.jsx >  Navbar >  handleAddClick
1   import React from "react";
2   import { Link, NavLink, useNavigate } from "react-router-dom";
```

Part 2 - MySQL Persistence and Server-Side Sessions

In this part, you will apply the concepts learned in class to build a simple CRUD (Create, Read, Update, Delete) application using FastAPI and MySQL, using SQLAlchemy.

Database

Create MySQL database: <PREFIX>_rel

Table: your primary domain entity (auto-increment id, primary field, secondary field)

```
PS C:\Users\natal> & "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -p
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.41 MySQL Community Server - GPL

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE IF NOT EXISTS s0932_rel
-> ;
Query OK, 1 row affected (0.03 sec)

mysql> exit;
Bye
```

```
mysql> DESCRIBE vulnerabilities;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| id             | int           | NO   | PRI | NULL    | auto_increment |
| package_name   | varchar(255)  | NO   |     | NULL    |                |
| severity       | varchar(255)  | NO   |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Also create tables for authentication/session management:

- users (id, name, email (unique), password_hash)

```

data260-0932 > reports > hw04 > backend > app > models.py > Vul
1  from sqlalchemy import Column, Integer, String, DateTime, ForeignKey
2  from .database import Base
3  from sqlalchemy.sql import func
4  from sqlalchemy.orm import relationship
5
6  class User(Base):
7      __tablename__ = "users"
8
9      id = Column(Integer, primary_key=True, index=True, autoincrement=True)
10     name = Column(String(255), nullable=False)
11     email = Column(String(255), nullable=False, unique=True)
12     password_hash=Column(String(255), nullable=False)

```

- sessions (id (session token), user_id, created_at, expires_at)

```

15  class SessionToken(Base):
16      __tablename__ = "sessions"
17
18      id = Column(String(64), primary_key=True, index=True) # session token
19      user_id = Column(Integer, ForeignKey("users.id", ondelete="CASCADE"), nullable=False)
20      created_at = Column(DateTime(timezone=True), server_default=func.now())
21      expires_at = Column(DateTime(timezone=True), nullable=False)

```

You are required to create a CRUD system for your domain entity that allows users to:

The screenshot shows the Postman API client interface. The top bar includes navigation links (Home, Workspaces, Explore), a search bar, and user options (Sign In, Create Account). A notification banner states: "You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman." The left sidebar displays a history of requests, categorized by "Today" and "Yesterday". The main panel shows a selected POST request to `http://localhost:8032/vuls`. The request body is a JSON object: `{"email": "t@gmail.com", "password": "test"}`. The response status is `200 OK` with a response time of `16 ms` and a body size of `295 B`. The response body is a JSON object: `{"message": "logged in", "user_id": 4}`.

- Add a new record (POST)

```
data260-0932 > reports > hw04 > backend > app > crud.py > update_user
44
45 #functions for vulnerabilities, follows same logic as useres demo
46 def create_vul(db: Session, payload: schema.VulCreate):
47     vul = models.Vul(package_name=payload.package_name, severity=payload.severity)
48     db.add(vul)
49     db.commit()
50     db.refresh(vul)
51     return vul
```

The screenshot shows the Postman interface. The top bar includes navigation links (Home, Workspaces, Explore), a search bar, and user options (Sign In, Create Account). A notification banner states: "You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman."

The left sidebar displays the "History" tab with a list of recent requests, categorized by "Today" and "Yesterday". The main workspace shows a POST request to `http://localhost:8032/vuls`. The "Body" tab is selected, showing a JSON payload:

```
1 {
2   "package_name": "python 3.11",
3   "severity": "Medium"
4 }
```

Below the request editor, the response is displayed in the "Body" tab, showing a 200 OK status with a response time of 15 ms and a body size of 182 B. The response is formatted as JSON:

```
1 {
2   "id": 2,
3   "package_name": "python 3.11",
4   "severity": "Medium"
5 }
```

Open Source Package Vulnerability Manager

Home Add Vulnerability Record Logout

Vulnerabilities

Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session.

+ Add Record

ID	PACKAGE NAME	SEVERITY	ACTIONS
2	python 3.11	Medium	Update Delete

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Method	Domain	File	Initiator	Type	Transferred	Size
200	GET	localhost:5173	client	script	js	137.97 kB	137.73 kB
200	GET	localhost:5173	main.jsx	script	js	3.09 kB	2.85 kB
200	GET	localhost:5173	@react-refresh	script	js	117.79 kB	117.55 kB
200	GET	localhost:5173	react_jsx-dev-runtime.js?v=0c4335a2	script	js	37.29 kB	37.04 kB
200	GET	localhost:5173	react.js?v=0c4335a2	script	js	473 B	219 B
200	GET	localhost:5173	react-dom_client.js?v=0c4335a2	script	js	1.37 kB	1.12 kB
200	GET	localhost:5173	react-router-dom.js?v=0c4335a2	script	js	210.25 kB	209.99 kB
200	GET	localhost:5173	app.jsx	script	js	19.80 kB	19.56 kB
200	GET	localhost:5173	styles.css	script	js	7.78 kB	7.55 kB
200	GET	localhost:5173	env.mjs	script	js	4.26 kB	4.02 kB
200	GET	localhost:5173	chunk-DRWLMN53.js?v=0c4335a2	script	js	78.08 kB	77.82 kB
200	GET	localhost:5173	chunk-G3PMV62Z.js?v=0c4335a2	script	js	1.80 kB	1.54 kB
200	GET	localhost:5173	chunk-PJEEZAML.js?v=0c4335a2	script	js	928.36 kB	928.10 kB
200	GET	localhost:5173	Navbar.jsx	script	js	10.33 kB	10.09 kB
200	GET	localhost:5173	Home.jsx	script	js	18.95 kB	18.71 kB
200	GET	localhost:5173	CreateRecord.jsx	script	js	10.28 kB	10.04 kB
200	GET	localhost:5173	UpdateRecord.jsx	script	js	12.55 kB	12.31 kB
200	GET	localhost:5173	DeleteRecord.jsx	script	js	10.12 kB	9.68 kB
200	GET	localhost:5173	Login.jsx	script	js	12.18 kB	11.94 kB
200	GET	localhost:5173	vulApi.js	script	js	6.07 kB	5.83 kB
101	GET	localhost:5173	/?token=7OVzptVjUfIt	client:536 (webso...	plain	163 B	0 B
200	GET	localhost:5173	axios.js?v=0c4335a2	script	js	85.73 kB	85.48 kB
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B
404	GET	localhost:5173	favicon.ico	img	plain	143 B	0 B
200	GET	localhost:8032	vuls	axios.js:1728 (xhr)	json	290 B	59 B

Filter properties

JSON

Raw

0: { id: 2, package_name: "python 3.11", severity: "Medium" }

- View all records (GET)

```
def get_vuls(db: Session):
    return db.query(models.Vul).order_by(models.Vul.id.asc()).all()
```

HomeWorkspacesExplore

Search Postman

Sign InCreate Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

NewImport

POSThttp://localhost:8032/vulsGEThttp://localhost:8032/vuls

POSThttp://localhost:8032/vulsGEThttp://localhost:8032/vuls

POSThttp://localhost:8032/vulsGEThttp://localhost:8032/vuls

POSThttp://localhost:8032/auth/login

POSThttp://localhost:8032/auth/register

POSThttp://localhost:8032/auth/register

POSThttp://localhost:8032/vuls

POSThttp://localhost:8032/vuls

POSThttp://localhost:8032/auth/records

POSThttp://localhost:8032/auth/records

POSThttp://localhost:8032/auth/login

POSThttp://localhost:8032/auth/login

POSThttp://localhost:8032/auth/register

POSThttp://localhost:8032/auth/register

POSThttp://localhost:8032/auth/register

POSThttp://localhost:8032/auth/register

POSThttp://localhost:8032/auth

POSThttp://localhost:8032/auth/register

POSThttp://localhost:8032/auth/register

POSThttp://localhost:8000/auth/register

POSThttp://localhost:8000/auth/register

POSThttp://localhost:8000/auth/register

POSThttp://localhost:8000/auth/register

POSThttp://localhost:8000/auth/register

GEThttp://localhost:8000/auth/register

http://localhost:8032/vuls

GEThttp://localhost:8032/vuls

Send

ParamsAuthorizationHeaders (9)BodyPre-request ScriptTestsSettingsCookiesBeautify

noneform-datax-www-form-urlencodedrawbinaryJSON

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BodyCookies (1)Headers (4)Test Results200 OK10 ms240 BSave Response

PrettyRawPreviewVisualizeJSON

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```

The screenshot shows a web application titled "Open Source Package Vulnerability Manager". The main section, "Vulnerabilities", displays a table of vulnerabilities. Below the table, a network inspector is open, showing a list of HTTP requests. The response for the request to "/token=7OVzptVjUfft" is highlighted, showing a JSON object with two records.

Vulnerabilities Table:

ID	PACKAGE NAME	SEVERITY	ACTIONS
2	python 3.11	Medium	Update Delete
3	java 2.3	Medium	Update Delete

Network Inspector:

Status	Meth...	Domain	File	Initiator	Type	Transferred	Size	Headers	Cookies	Request	Response	Timings	Stack Trace
200	GET	localhost:5173	client	script	js	137.97 kB	137.7...						
200	GET	localhost:5173	main.jsx	script	js	3.09 kB	2.85 kB						
200	GET	localhost:5173	@react-refresh	script	js	117.79 kB	117.5...						
200	GET	localhost:5173	react_jsx-dev-runtime.js?v=0c4335a2	script	js	37.29 kB	37.04...						
200	GET	localhost:5173	react.js?v=0c4335a2	script	js	473 B	219 B						
200	GET	localhost:5173	react-dom_client.js?v=0c4335a2	script	js	1.37 kB	1.12 kB						
200	GET	localhost:5173	react-router-dom.js?v=0c4335a2	script	js	210.25 kB	209.9...						
200	GET	localhost:5173	app.jsx	script	js	19.80 kB	19.56...						
200	GET	localhost:5173	styles.css	script	js	7.78 kB	7.55 kB						
200	GET	localhost:5173	env.mjs	script	js	4.26 kB	4.02 kB						
200	GET	localhost:5173	chunk-PJEEZAML.js?v=0c4335a2	script	js	928.36 kB	928.1...						
200	GET	localhost:5173	chunk-DRWLMN53.js?v=0c4335a2	script	js	78.08 kB	77.82...						
200	GET	localhost:5173	chunk-G3PMV62Z.js?v=0c4335a2	script	js	1.80 kB	1.54 kB						
200	GET	localhost:5173	Navbar.jsx	script	js	10.33 kB	10.09...						
200	GET	localhost:5173	Home.jsx	script	js	18.95 kB	18.71...						
200	GET	localhost:5173	CreateRecord.jsx	script	js	10.28 kB	10.04...						
200	GET	localhost:5173	UpdateRecord.jsx	script	js	12.55 kB	12.31...						
200	GET	localhost:5173	DeleteRecord.jsx	script	js	10.12 kB	9.88 kB						
200	GET	localhost:5173	Login.jsx	script	js	12.18 kB	11.94...						
200	GET	localhost:5173	vulApi.js	script	js	6.07 kB	5.83 kB						
401	GET	localhost:5173	/token=7OVzptVjUfft	client536 (webs...	plain	163 B	0 B						
200	GET	localhost:5173	axios.js?v=0c4335a2	script	js	85.73 kB	85.48...						
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B						
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B						
404	GET	localhost:5173	favicon.ico	img	plain	143 B	0 B						
200	GET	localhost:8032	vuls	axios.js:1728 (xhr)	json	346 B	114 B						

- View a record by ID (GET)

```
def get_vul(db: Session, vul_id: int):
    return db.query(models.Vul).filter(models.Vul.id == vul_id).first()
```

Home Workspaces Explore Search Postman Sign In Create Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History New Import POST http://localhost:8032/vul GET http://localhost:8032/vul

Today

- GET http://localhost:8032/vuls/3
- GET http://localhost:8032/vuls
- POST http://localhost:8032/vuls
- POST http://localhost:8032/vuls
- POST http://localhost:8032/auth/login
- POST http://localhost:8032/auth/register
- POST http://localhost:8032/auth/register
- POST http://localhost:8032/vuls
- POST http://localhost:8032/vuls
- POST http://localhost:8032/auth/records
- POST http://localhost:8032/auth/records
- POST http://localhost:8032/auth/login
- POST http://localhost:8032/auth/login

http://localhost:8032/vuls Save </>

GET http://localhost:8032/vuls/3 Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

Query Params

Key	Value	Bulk Edit
Key	Value	

Body Cookies (1) Headers (4) Test Results 200 OK 12 ms 179 B Save Response

Pretty Raw Preview Visualize JSON

```

1  {
2    "id": 3,
3    "package_name": "java 2.3",
4    "severity": "Medium"
5  }

```

• Update record details (PUT)

```

def update_vul(db: Session, vul_id: int, payload: schema.VulUpdate):
    vul = get_vul(db, vul_id)
    if not vul:
        return None
    vul.package_name = payload.package_name
    vul.severity = payload.severity
    db.commit()
    db.refresh(vul)
    return vul

```

History New Import POST http://localhost:8032/vul PUT http://localhost:8032/vul

Today

- PUT http://localhost:8032/vuls/2
- GET http://localhost:8032/vuls/3
- GET http://localhost:8032/vuls
- POST http://localhost:8032/vuls
- POST http://localhost:8032/vuls
- POST http://localhost:8032/auth/login
- POST http://localhost:8032/auth/register
- POST http://localhost:8032/auth/register
- POST http://localhost:8032/vuls
- POST http://localhost:8032/vuls
- POST http://localhost:8032/auth/records
- POST http://localhost:8032/auth/records
- POST http://localhost:8032/auth/login
- POST http://localhost:8032/auth/login
- POST http://localhost:8032/auth/register

http://localhost:8032/vuls Save </>

PUT http://localhost:8032/vuls/2 Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies Beautify

none form-data x-www-form-urlencoded raw binary JSON

```

1  {
2    "package_name": "python 3.14",
3    "severity": "Critical"
4  }
5

```

Body Cookies (1) Headers (4) Test Results 200 OK 16 ms 184 B Save Response

Pretty Raw Preview Visualize JSON

```

1  {
2    "id": 2,
3    "package_name": "python 3.14",
4    "severity": "Critical"
5  }

```

Open Source Package Vulnerability Manager

Home Add Vulnerability Record Logout

Vulnerabilities

Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session.

+ Add Record

ID	PACKAGE NAME	SEVERITY	ACTIONS
2	python 3.14	Critical	Update Delete
3	java 2.3	Medium	Update Delete

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Filter URLs

Status	Meth...	Domain	File	Initiator	Type	Transferred	Size	Headers	Cookies	Request	Response	Timings	Stack Trace
200	GET	localhost:5173	client	script	js	137.97 kB	137.7...						
200	GET	localhost:5173	main.jsx	script	js	3.09 kB	2.85 kB						
200	GET	localhost:5173	@react-refresh	script	js	117.79 kB	117.5...						
200	GET	localhost:5173	react_jsx-dev-runtime.js?v=0c4335a2	script	js	37.29 kB	37.04 ...						
200	GET	localhost:5173	react.js?v=0c4335a2	script	js	473 B	219 B						
200	GET	localhost:5173	react-dom_client.js?v=0c4335a2	script	js	1.37 kB	1.12 kB						
200	GET	localhost:5173	react-router-dom.js?v=0c4335a2	script	js	210.25 kB	209.9...						
200	GET	localhost:5173	app.jsx	script	js	19.80 kB	19.56 ...						
200	GET	localhost:5173	styles.css	script	js	7.78 kB	7.55 kB						
200	GET	localhost:5173	env.mjs	script	js	4.26 kB	4.02 kB						
200	GET	localhost:5173	chunk-PJEEZAML.js?v=0c4335a2	script	js	928.36 kB	928.1...						
200	GET	localhost:5173	chunk-DRWLMN53.js?v=0c4335a2	script	js	78.06 kB	77.82 ...						
200	GET	localhost:5173	chunk-G3PMV62Z.js?v=0c4335a2	script	js	1.80 kB	1.54 kB						
200	GET	localhost:5173	Navbar.jsx	script	js	10.33 kB	10.09 ...						
200	GET	localhost:5173	Home.jsx	script	js	18.95 kB	18.71 ...						
200	GET	localhost:5173	CreateRecord.jsx	script	js	10.28 kB	10.04 ...						
200	GET	localhost:5173	UpdateRecord.jsx	script	js	12.55 kB	12.31 ...						
200	GET	localhost:5173	DeleteRecord.jsx	script	js	10.12 kB	9.88 kB						
200	GET	localhost:5173	Login.jsx	script	js	12.18 kB	11.94 ...						
200	GET	localhost:5173	vulApi.js	script	js	6.07 kB	5.83 kB						
101	GET	localhost:5173	/?token=7OVzptVyuUft	client:536 (webs...	plain	163 B	0 B						
200	GET	localhost:5173	axios.js?v=0c4335a2	script	js	85.73 kB	85.48 ...						
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B						
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B						
200	GET	localhost:8032	vuls	axios.js:1728 (xhr)	json	348 B	116 B						

Filter properties

JSON

Raw

0: { id: 2, package_name: "python 3.14", severity: "Critical" }

1: { id: 3, package_name: "java 2.3", severity: "Medium" }

- Delete a record (DELETE)

```
def delete_vul(db: Session, vul_id: int):
    vul = get_vul(db, vul_id)
    if not vul:
        return None
    db.delete(vul)
    db.commit()
    return vul
```

HomeWorkspacesExplore

Search Postman

Sign InCreate Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

History

NewImport

POST http://localhost:8032/vulDEL http://localhost:8032/vul: + ...

Today

DEL http://localhost:8032/vuls/2

PUT http://localhost:8032/vuls/2

GET http://localhost:8032/vuls/3

GET http://localhost:8032/vuls

POST http://localhost:8032/vuls

POST http://localhost:8032/vuls

POST http://localhost:8032/auth/login

POST http://localhost:8032/auth/register

POST http://localhost:8032/auth/register

POST http://localhost:8032/vuls

POST http://localhost:8032/vuls

POST http://localhost:8032/auth/records

POST http://localhost:8032/auth/records

POST http://localhost:8032/auth/login

POST http://localhost:8032/auth/login

http://localhost:8032/vuls

Save</>

DELETE http://localhost:8032/vuls/2 Send

ParamsAuthorizationHeaders (9)BodyPre-request ScriptTestsSettingsCookies

Query Params

Key	Value	Bulk Edit
Key	Value	

BodyCookies (1)Headers (4)Test Results200 OK16 ms184 BSave Response

PrettyRawPreviewVisualizeJSON

1

2

3

4

5

id: 2,

package_name: "python 3.14",

severity: "Critical"

The screenshot shows a web browser at `http://localhost:5173` displaying the 'Open Source Package Vulnerability Manager' application. The page has a header with 'Home', 'Add Vulnerability Record', and 'Logout' links. Below the header is a 'Vulnerabilities' section with a '+ Add Record' button and a note: 'Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session.'

The main content area displays a table of vulnerabilities:

ID	PACKAGE NAME	SEVERITY	ACTIONS
3	java 2.3	Medium	<button>Update</button> <button>Delete</button>

Below the table is a network inspector showing a list of requests. The selected request is a GET request to `/token=7OVzpVvUff` with a status of 200. The response is a JSON object: `{ id: 3, package_name: 'java 2.3', severity: 'Medium' }`.

You are free to name your API endpoints appropriately.

The session token issued at login must be stored server-side in the sessions table and referenced only by an opaque token in the browser's HTTP-only cookie - the cookie itself should never store user data directly.

Open Source Package Vulnerability Manager

Home Add Vulnerability Record Logout

Login

Log in to view and manage vulnerability records

Email

Password

Login

Status	Meth...	Domain	File	Initiator	Type	Transferred	Size	Headers	Cookies	Request	Response	Timings
200	GET	localhost:5173	login	document	html	816 B	565 B	Filter Headers				
200	GET	localhost:5173	client	script	js	137.97 kB	137.7...					
200	GET	localhost:5173	main.js	script	js	3.09 kB	2.85 kB					
200	GET	localhost:5173	@react-refresh	script	js	117.79 kB	117.5...					
200	GET	localhost:5173	react_jsx-dev-runtime.js?v=0c4335a2	script	js	37.29 kB	37.04 ...					
200	GET	localhost:5173	react.js?v=0c4335a2	script	js	473 B	219 B					
200	GET	localhost:5173	react-dom_client.js?v=0c4335a2	script	js	1.37 kB	1.12 kB					
200	GET	localhost:5173	react-router-dom.js?v=0c4335a2	script	js	210.25 kB	209.9...					
200	GET	localhost:5173	app.js	script	js	19.80 kB	19.56 ...					
200	GET	localhost:5173	styles.css	script	js	7.78 kB	7.55 kB					
200	GET	localhost:5173	env.mjs	script	js	4.26 kB	4.02 kB					
200	GET	localhost:5173	chunk-DRWLMNS3.js?v=0c4335a2	script	js	78.08 kB	77.82 ...					
200	GET	localhost:5173	chunk-G3PMV62Z.js?v=0c4335a2	script	js	1.80 kB	1.54 kB					
200	GET	localhost:5173	chunk-PJEEZAML.js?v=0c4335a2	script	js	928.36 kB	928.1...					
200	GET	localhost:5173	Navbar.js	script	js	10.33 kB	10.09 ...					
200	GET	localhost:5173	Home.js	script	js	18.95 kB	18.71 ...					
200	GET	localhost:5173	CreateRecord.js	script	js	10.28 kB	10.04 ...					
200	GET	localhost:5173	UpdateRecord.js	script	js	12.55 kB	12.31 ...					
200	GET	localhost:5173	DeleteRecord.js	script	js	10.12 kB	9.88 kB					
200	GET	localhost:5173	Login.js	script	js	12.18 kB	11.94 ...					
200	GET	localhost:5173	vulApi.js	script	js	6.07 kB	5.83 kB					
101	GET	localhost:5173	/token=7OVzptVyUfft	client:536 (webs...	plain	163 B	0 B					
200	GET	localhost:5173	axios.js?v=0c4335a2	script	js	85.73 kB	85.48 ...					
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B					
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B					
200	GET	localhost:8032	vuls	axios.js:1728 (xhr)	json	287 B	56 B					

27 requests 1.71 MB / 1.73 MB transferred 4.15 s DOMContentLoaded: 3.05 s load: 3.18 s

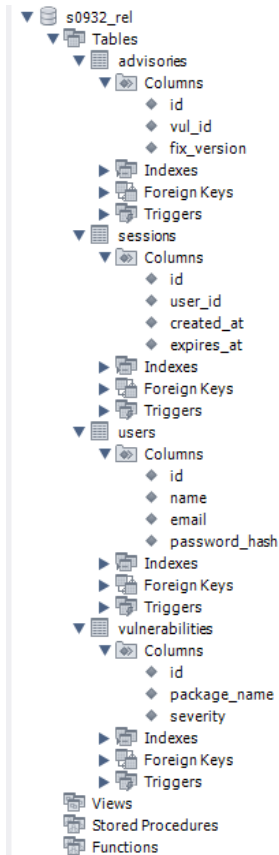
Response Headers (231 B)

- Cache-Control: no-cache
- Connection: keep-alive
- Content-Length: 585
- Content-Type: text/html
- Date: Mon, 28 Sep 2026 17:07:35 GMT
- Etag: W/"249-C7UZYLRWUUCZHy/qiSeRQ/oiRc"
- Keep-Alive: timeout=5
- Vary: Origin

Request Headers (593 B)

- Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
- Accept-Encoding: gzip, deflate, br, zstd
- Accept-Language: en-US,en;q=0.9
- Cache-Control: no-cache
- Connection: keep-alive
- Cookie: session_id=8a57433a8a42f4dc1efcb534e71e8543cd0ae66ac111db5d22bca4c3516ed5
- Host: localhost:5173
- Pragma: no-cache
- Priority: u=0,i
- Sec-Fetch-Dest: document
- Sec-Fetch-Mode: navigate
- Sec-Fetch-Site: none
- Sec-Fetch-User: ?1

Additionally submit Postman API response screenshots for each operation. Also submit a screenshot of your database and your project folder structure



Part 3 - N+1 Measurement and Query Tuning

1 Seed 5,000 rows of your primary domain entity and 200 related rows, using your SEED value. This related table is just test data for now - you'll build a proper related entity with full CRUD in later homework. Commit your generator script and the schema/migration you used to create these rows.

```

data260-0932 > reports > hw04 > backend > seed.py > ...
1  import random
2  from sqlalchemy import text
3  from app.database import SessionLocal, engine, Base
4  from app import models
5
6  random.seed(932)
7
8
9  Base.metadata.drop_all(bind=engine)
10 Base.metadata.create_all(bind=engine)
11 db = SessionLocal()
12
13 db.query(models.Advisory).delete()
14 db.query(models.Vul).delete()
15 # make the ids start at 1 again
16 db.execute(text("ALTER TABLE vulnerabilities AUTO_INCREMENT = 1"))
17 db.execute(text("ALTER TABLE advisories AUTO_INCREMENT = 1"))
18 db.commit()
19
20 names = ["lodash", "express", "axios", "requests", "django", "flask", "numpy", "react", "vue",
21 severities = ["Low", "Medium", "High", "Critical"]
22
23 vulns = []
24 for i in range(5000):
25     vulns.append(models.Vul(
26         package_name=random.choice(names) + "-" + str(i),
27         severity=random.choice(severities),))
28
29 db.add_all(vulns)
30 db.commit()
31
32 ids = [row.id for row in db.query(models.Vul.id).order_by(models.Vul.id).all()]
33
34 advisories = []
35 for i in range(200):
36     fix = str(random.randint(1, 9)) + "." + str(random.randint(0, 9)) + "." + str(random.randint(0, 9))
37     advisories.append(models.Advisory(vul_id=random.choice(ids[:200]), fix_version=fix))
38 db.add_all(advisories)
39 db.commit()
40
41 print("Seeded", len(vulns), "vulnerabilities and", len(advisories), "advisories (SEED = 932)")
42

```

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04\backend> python .\seed.py
Seeded 5000 vulnerabilities and 200 advisories (SEED = 932)

```

2 The 200 related rows must be associated with the primary domain entity, and the list endpoint's response must include this related data.

Open Source Package Vulnerability Manager

HomeAdd Vulnerability RecordLogout

Vulnerabilities

Session-based access: these open source vulnerabilities are fetched from FastAPI + MySQL using your cookie session.

+ Add Record

ID	PACKAGE NAME	SEVERITY	ACTIONS
1	vue-0	Low	UpdateDelete
2	vue-1	Critical	UpdateDelete

InspectorConsoleDebuggerNetworkStyle EditorPerformanceMemoryStorageAccessibilityApplication

Filter URLs

HTMLCSSJSXHRFontsImagesMediaWSOtherDisable CacheNo Throttling

Status	Meth...	Domain	File	Initiator	Type	Transferred	Size
200	GET	localhost:5173	client	script	js	137.97 kB	137.7...
200	GET	localhost:5173	main.jsx	script	js	3.09 kB	2.85 kB
200	GET	localhost:5173	@react-refresh	script	js	117.79 kB	117.5...
200	GET	localhost:5173	react_jsx-dev-runtime.js?v=0c4335a2	script	js	37.29 kB	37.04 ...
200	GET	localhost:5173	react.js?v=0c4335a2	script	js	473 B	219 B
200	GET	localhost:5173	react-dom_client.js?v=0c4335a2	script	js	1.37 kB	1.12 kB
200	GET	localhost:5173	react-router-dom.js?v=0c4335a2	script	js	210.25 kB	209.9...
200	GET	localhost:5173	app.jsx	script	js	19.80 kB	19.56 ...
200	GET	localhost:5173	styles.css	script	js	7.78 kB	7.55 kB
200	GET	localhost:5173	env.mjs	script	js	4.26 kB	4.02 kB
200	GET	localhost:5173	chunk-PJEEZAML.js?v=0c4335a2	script	js	928.36 kB	928.1...
200	GET	localhost:5173	chunk-DRWLMN53.js?v=0c4335a2	script	js	78.08 kB	77.82 ...
200	GET	localhost:5173	chunk-G3PMV62Z.js?v=0c4335a2	script	js	1.80 kB	1.54 kB
200	GET	localhost:5173	Navbar.jsx	script	js	10.33 kB	10.09 ...
200	GET	localhost:5173	Home.jsx	script	js	18.95 kB	18.71 ...
200	GET	localhost:5173	CreateRecord.jsx	script	js	10.28 kB	10.04 ...
200	GET	localhost:5173	UpdateRecord.jsx	script	js	12.55 kB	12.31 ...
200	GET	localhost:5173	DeleteRecord.jsx	script	js	10.12 kB	9.88 kB
200	GET	localhost:5173	Login.jsx	script	js	12.18 kB	11.94 ...
200	GET	localhost:5173	vulApi.js	script	js	6.07 kB	5.83 kB
101	GET	localhost:5173	?token=58zq8RgPaWA-	client536 (webs...	plain	163 B	0 B
200	GET	localhost:5173	axios.js?v=0c4335a2	script	js	85.73 kB	85.48 ...
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B
200	GET	localhost:8032	me	axios.js:1728 (xhr)	json	261 B	30 B
404	GET	localhost:5173	favicon.ico	img	plain	143 B	0 B
200	GET	localhost:8032	vuls	axios.js:1728 (xhr)	json	297.29 kB	297.0...

27 requests2.01 MB / 2.01 MB transferredFinish: 4.35 sDOMContentLoaded: 2.05 sInit: 2.28 s

Filter propertiesJSONRaw

[{"id": 4970, "package_name": "vue-4970", "severity": "Medium"}, {"id": 4971, "package_name": "requests-4971", "severity": "Low"}, {"id": 4972, "package_name": "numpy-4972", "severity": "High"}, {"id": 4973, "package_name": "react-4973", "severity": "Medium"}, {"id": 4974, "package_name": "flask-4974", "severity": "High"}, {"id": 4975, "package_name": "axios-4975", "severity": "Medium"}, {"id": 4976, "package_name": "jquery-4976", "severity": "High"}, {"id": 4977, "package_name": "vue-4977", "severity": "Critical"}, {"id": 4978, "package_name": "axios-4978", "severity": "Critical"}, {"id": 4979, "package_name": "flask-4979", "severity": "Critical"}, {"id": 4980, "package_name": "vue-4980", "severity": "Critical"}, {"id": 4981, "package_name": "django-4981", "severity": "Low"}, {"id": 4982, "package_name": "flask-4982", "severity": "High"}, {"id": 4983, "package_name": "jquery-4983", "severity": "Medium"}, {"id": 4984, "package_name": "axios-4984", "severity": "Critical"}, {"id": 4985, "package_name": "lodash-4985", "severity": "High"}, {"id": 4986, "package_name": "lodash-4986", "severity": "Low"}, {"id": 4987, "package_name": "react-4987", "severity": "Low"}, {"id": 4988, "package_name": "numpy-4988", "severity": "Medium"}, {"id": 4989, "package_name": "react-4989", "severity": "Medium"}, {"id": 4990, "package_name": "requests-4990", "severity": "Critical"}, {"id": 4991, "package_name": "axios-4991", "severity": "High"}, {"id": 4992, "package_name": "vue-4992", "severity": "Critical"}, {"id": 4993, "package_name": "flask-4993", "severity": "Critical"}, {"id": 4994, "package_name": "react-4994", "severity": "Low"}, {"id": 4995, "package_name": "requests-4995", "severity": "Low"}, {"id": 4996, "package_name": "flask-4996", "severity": "Medium"}, {"id": 4997, "package_name": "vue-4997", "severity": "Low"}, {"id": 4998, "package_name": "requests-4998", "severity": "Medium"}]

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04\backend>python .\seed.py
Seeded 5000 vulnerabilities and 200 advisories (SEED = 932)

3 Build a list endpoint that runs one extra query per record to fetch its related data (this is the classic "N+1" problem - intentionally naive for now).

```
@app.get("/vuls/naive", response_model=list[Schema.VulAdvi])
def list_naive(response: Response, limit: int = 10, db: Session = Depends(get_db), _session = Depends(require_session)):
    query_count["n"] = 0
    vulns = db.query(models.Vul).order_by(models.Vul.id).limit(limit).all()
    result = []
    for v in vulns:
        # the extra query, run once per record
        advs = db.query(models.Advisory).filter(models.Advisory.vul_id == v.id).all()
        result.append({"id": v.id, "package_name": v.package_name, "severity": v.severity, "advisories": advs})
    response.headers["X-SQL-Count"] = str(query_count["n"])
    return result
```

Response Headers

Permissions-Policy join-ad-interest-group=(), run-ad-auction=(), browsing-topics=(), private-state-token-issuance=(), private-state-token-redemption=()

content-length 879

content-type application/json

date Mon, 28 Sep 2026 17:38:54 GMT

server uvicorn

x-sql-count 11

Request Headers

Accept text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8

Accept-Encoding gzip, deflate, br, zstd

Accept-Language en-US,en;q=0.9

Connection keep-alive

Host localhost:8032

Priority u=0, i

Sec-Fetch-Dest document

Sec-Fetch-Mode navigate

Sec-Fetch-Site none

Sec-Fetch-User ?1

Sec-GPC 1

Upgrade-Insecure-Requests 1

User-Agent Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:156.0) Gecko/20100101 Firefox/156.0

4 Test this endpoint at page sizes 10, 50, and 200. For each size, run 30 requests and record how many SQL queries each request triggers, plus the p50/p95/p99 latency.

```
print(time.strftime("%Y-%m-%d %H:%M:%S"), "start measuring")
for size in SIZES:
    for version in ["naive", "fixed"]:
        times = []
        counts = []
        for n in range(1, 31):
            start = time.perf_counter()
            r = s.get(BASE + "/vuls/" + version, params={"limit": size})
            ms = (time.perf_counter() - start) * 1000
            sql_count = int(r.headers["X-SQL-Count"])
            times.append(ms)
            counts.append(sql_count)
            raw_rows.append([size, version, n, r.status_code, sql_count, round(ms, 2)])

p50, p95, p99 = percentile(times, 50), percentile(times, 95), percentile(times, 99)
summary_rows.append([size, version, round(p50, 2), round(p95, 2), round(p99, 2), counts[0]])
print(time.strftime("%Y-%m-%d %H:%M:%S"), "size", size, version, "p50", round(p50, 2), "p95", round(p95, 2), "p99", round(p99, 2), "stmts/req", counts[0])
```

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04\backend> python .\measure.py
2026-09-28 10:20:05 start measuring
2026-09-28 10:20:05 size 10 naive p50 5.85 p95 6.5 p99 7.07 stmts/req 11
2026-09-28 10:20:05 size 10 fixed p50 3.06 p95 3.64 p99 4.96 stmts/req 1
2026-09-28 10:20:06 size 50 naive p50 17.3 p95 18.33 p99 18.54 stmts/req 51
2026-09-28 10:20:06 size 50 fixed p50 3.86 p95 4.64 p99 22.06 stmts/req 1
2026-09-28 10:20:08 size 200 naive p50 60.85 p95 66.89 p99 77.59 stmts/req 201
2026-09-28 10:20:08 size 200 fixed p50 6.06 p95 23.67 p99 25.55 stmts/req 1
2026-09-28 10:20:08 done, saved 180 requests to .. /raw/
```

5 Now fix the N+1 issue using a join or eager loading, and re-run the same Measurements.

```

print(time.strftime("%Y-%m-%d %H:%M:%S"), "start measuring")
for size in SIZES:
    for version in ["naive", "fixed"]:
        times = []
        counts = []
        for n in range(1, 31):
            start = time.perf_counter()
            r = s.get(BASE + "/vuls/" + version, params={"limit": size})
            ms = (time.perf_counter() - start) * 1000
            sql_count = int(r.headers["X-SQL-Count"])
            times.append(ms)
            counts.append(sql_count)
            raw_rows.append([size, version, n, r.status_code, sql_count, round(ms, 2)])

p50, p95, p99 = percentile(times, 50), percentile(times, 95), percentile(times, 99)
summary_rows.append([size, version, round(p50, 2), round(p95, 2), round(p99, 2), counts[0]
print(time.strftime("%Y-%m-%d %H:%M:%S"), "size", size, version, "p50", round(p50, 2), "p95", round(p95, 2), "p99", round(p99, 2), "stmts/req", counts[0])

```

```

@app.get("/vuls/fixed", response_model=list[schema.VulAdvi])
def list_fixed(response: Response, limit: int = 10, db: Session = Depends(get_db), _session = Depends(require_session)):
    query_count["n"]=0
    vulns = (db.query(models.Vul)
              .options(joinedload(models.Vul.advisories))
              .order_by(models.Vul.id).limit(limit).all())
    result = []
    for v in vulns:
        result.append({"id": v.id, "package_name": v.package_name, "severity": v.severity, "advisories": v.advisories})
    response.headers["X-SQL-Count"] = str(query_count["n"])
    return result

```

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04\backend> python .\measure.py
2026-09-28 10:20:05 start measuring
2026-09-28 10:20:05 size 10 naive p50 5.85 p95 6.5 p99 7.07 stmts/req 11
2026-09-28 10:20:05 size 10 fixed p50 3.06 p95 3.64 p99 4.96 stmts/req 1
2026-09-28 10:20:06 size 50 naive p50 17.3 p95 18.33 p99 18.54 stmts/req 51
2026-09-28 10:20:06 size 50 fixed p50 3.86 p95 4.64 p99 22.06 stmts/req 1
2026-09-28 10:20:08 size 200 naive p50 60.85 p95 66.89 p99 77.59 stmts/req 201
2026-09-28 10:20:08 size 200 fixed p50 6.06 p95 23.67 p99 25.55 stmts/req 1
2026-09-28 10:20:08 done, saved 180 requests to ../raw/

```

6 Save the raw data for all 180 requests (3 page sizes × 2 versions × 30 requests) in reports/hw04/raw/, so your results can be double-checked.

```

print(time.strftime("%Y-%m-%d %H:%M:%S"), "start measuring")
for size in SIZES:
    for version in ["naive", "fixed"]:
        times = []
        counts = []
        for n in range(1, 31):
            start = time.perf_counter()
            r = s.get(BASE + "/vuls/" + version, params={"limit": size})
            ms = (time.perf_counter() - start) * 1000
            sql_count = int(r.headers["X-SQL-Count"])
            times.append(ms)
            counts.append(sql_count)
            raw_rows.append([size, version, n, r.status_code, sql_count, round(ms, 2)])

p50, p95, p99 = percentile(times, 50), percentile(times, 95), percentile(times, 99)
summary_rows.append([size, version, round(p50, 2), round(p95, 2), round(p99, 2), counts[0]
print(time.strftime("%Y-%m-%d %H:%M:%S"), "size", size, version, "p50", round(p50, 2), "p95", round(p95, 2), "p99", round(p99, 2), "stmts/req", counts[0])

```

```

with open("../raw/n_plus_one_raw.csv", "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(["page_size", "version", "request_no", "status", "sql_statements", "latency_ms"])
    w.writerows(raw_rows)

```

```

data260-0932 > reports > hw04 > raw > n_plus_one_raw.csv > data
1  page_size,version,request_no,status,sql_statements,latency_ms
2  10,naive,1,200,11,7.07
3  10,naive,2,200,11,6.47
4  10,naive,3,200,11,6.02
5  10,naive,4,200,11,6.18
6  10,naive,5,200,11,6.1
7  10,naive,6,200,11,5.49
8  10,naive,7,200,11,6.16
9  10,naive,8,200,11,5.85
10 10,naive,9,200,11,5.71
11 10,naive,10,200,11,6.02
12 10,naive,11,200,11,5.88
13 10,naive,12,200,11,5.65
14 10,naive,13,200,11,5.7
15 10,naive,14,200,11,6.5
16 10,naive,15,200,11,6.3
17 10,naive,16,200,11,5.38
18 10,naive,17,200,11,5.64
19 10,naive,18,200,11,5.73

```

7 Report how much faster the fixed version is at each page size, and explain why the speed-up changes as the page size grows.

```

print(time.strftime("%Y-%m-%d %H:%M:%S"), "start measuring")
for size in SIZES:
    for version in ["naive", "fixed"]:
        times = []
        counts = []
        for n in range(1, 31):
            start = time.perf_counter()
            r = s.get(BASE + "/vuls/" + version, params={"limit": size})
            ms = (time.perf_counter() - start) * 1000
            sql_count = int(r.headers["X-SQL-Count"])
            times.append(ms)
            counts.append(sql_count)
            raw_rows.append([size, version, n, r.status_code, sql_count, round(ms, 2)])

p50, p95, p99 = percentile(times, 50), percentile(times, 95), percentile(times, 99)
summary_rows.append([size, version, round(p50, 2), round(p95, 2), round(p99, 2), counts[0]])
print(time.strftime("%Y-%m-%d %H:%M:%S"), "size", size, version, "p50", round(p50, 2), "p95", round(p95, 2), "p99", round(p99, 2), "stmts/req", counts[0])

with open("../raw/n_plus_one_summary.csv", "w", newline="") as f:
    w = csv.writer(f)
    w.writerow(["page_size", "version", "p50_ms", "p95_ms", "p99_ms", "stmts_per_req"])
    w.writerows(summary_rows)

```

```

data260-0932 > reports > hw04 > raw > n_plus_one_summary.csv > data
1  page_size,version,p50_ms,p95_ms,p99_ms,stmts_per_req
2  10,naive,5.85,6.5,7.07,11
3  10,fixed,3.06,3.64,4.96,1
4  50,naive,17.3,18.33,18.54,51
5  50,fixed,3.86,4.64,22.06,1
6  200,naive,60.85,66.89,77.59,201
7  200,fixed,6.06,23.67,25.55,1
8

```

The speed up changes as the page size grows because the naive adds one query per record. One query is done to fetch the vulnerabilities and another to fetch the advisors for one page record. The fix only does one query in total to get both results. Therefore, the speed up changes when the page size increases.

8 Add one index to your database, and show the EXPLAIN output before and after adding it - explain what changed.

Adding the index changed how MySQL found the matching rows. It scanned the whole table before and then filtered to the specific vul_id value. Now, it uses a ref query type to find the specific index and match the rows. This is more efficient because only one row was checked.

```

data260-0932 > reports > hw04 > backend > explain_index.py > ...
1 from sqlalchemy import text
2 from app.database import engine
3
4 query = "EXPLAIN SELECT * FROM advisories WHERE vul_id = 42"
5 index_name = "idx_advisories_vul_id"
6
7 def run_explain(conn):
8     result = conn.execute(text(query))
9     lines = [" | ".join(result.keys())]
10    for row in result.fetchall():
11        lines.append(" | ".join(str(x) for x in row))
12    return "\n".join(lines)
13
14 with engine.connect() as conn:
15     try:
16         conn.execute(text("DROP INDEX " + index_name + " ON advisories"))
17         conn.commit()
18     except Exception:
19         conn.rollback()
20
21     before = run_explain(conn)
22
23     # add the one index
24     conn.execute(text("CREATE INDEX " + index_name + " ON advisories (vul_id)"))
25     conn.commit()
26
27     after = run_explain(conn)
28
29 print(query)
30 print(" --- BEFORE (no index) --- ")
31 print(before)
32 print(" --- AFTER (index added) --- ")
33 print(after)
34
35 with open("../raw/explain_before.txt", "w") as f:
36     f.write(query + "\n" + before + "\n")
37 with open("../raw/explain_after.txt", "w") as f:
38     f.write(query + "\n" + after + "\n")

```

```

PS C:\Users\natal\Documents\masters\fall_26\data_260\github\data260-0932\reports\hw04\backend> python .\explain_index.py
EXPLAIN SELECT * FROM advisories WHERE vul_id = 42
--- BEFORE (no index) ---
id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra
1 | SIMPLE | advisories | None | ALL | None | None | None | None | 200 | 10.0 | Using where
--- AFTER (index added) ---
id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra
1 | SIMPLE | advisories | None | ref | idx_advisories_vul_id | idx_advisories_vul_id | 4 | const | 1 | 100.0 | None

```

```

data260-0932 > reports > hw04 > raw > explain_after.txt
1 EXPLAIN SELECT * FROM advisories WHERE vul_id = 42
2 id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra
3 1 | SIMPLE | advisories | None | ref | idx_advisories_vul_id | idx_advisories_vul_id | 4 | const | 1 | 100.0 | None
4

```

```

data260-0932 > reports > hw04 > raw > explain_before.txt
1 EXPLAIN SELECT * FROM advisories WHERE vul_id = 42
2 id | select_type | table | partitions | type | possible_keys | key | key_len | ref | rows | filtered | Extra
3 1 | SIMPLE | advisories | None | ALL | None | None | None | None | 200 | 10.0 | Using where
4

```

9 Include Postman screenshots showing both versions of the endpoint working correctly at each page size.

Naive size 10

HomeWorkspacesExplore

Search Postman

Sign InCreate Account

You are using the Lightweight API Client, sign in or create an account to work with collections, environments and unlock all free features in Postman.

HistoryNewImport

POST http://localhost:8032/vul

GET http://localhost:8032/vul

POST http://localhost:8032/vul

POST http://localhost:8032/vul

GET http://localhost:8032/vul

+

...

Today

GET http://localhost:8032/vuls/naive?limit=10

GET http://localhost:8032/vulnerabilities/naive?limit=10

GET http://localhost:8032/vulnerabilities/naive?limit=10

POST GET http://localhost:8032/vulnerabilities/naive?limit=10

POST http://localhost:8032/auth/login

POST http://localhost:8032/auth/register

POST http://localhost:8032/auth/login

POST http://localhost:8032/auth/login

GET GET http://localhost:8032/vulnerabilities/naive?limit=10

DEL GET http://localhost:8032/vulnerabilities/naive?limit=10

DEL http://localhost:8032/vuls/2

PUT http://localhost:8032/vuls/2

GET http://localhost:8032/vuls/3

GET http://localhost:8032/vuls

POST http://localhost:8032/vuls

POST http://localhost:8032/vuls

POST http://localhost:8032/auth/login

POST http://localhost:8032/auth/register

POST http://localhost:8032/auth/register

POST http://localhost:8032/vuls

POST http://localhost:8032/auth/records

POST http://localhost:8032/auth/records

POST http://localhost:8032/auth/login

POST http://localhost:8032/auth/login

POST http://localhost:8032/auth/register

POST http://localhost:8032/auth/register

POST http://localhost:8032/auth/register

POST http://localhost:8032/auth/register

POST http://localhost:8032/auth/register

POST http://localhost:8000/auth/register

Yesterday

POST http://localhost:8000/auth/register

POST http://localhost:8000/auth/register

POST http://localhost:8000/auth/register

Create collections in Postman

Use collections to save your requests and share them with others.

Create a Collection

http://localhost:8032/auth/login

GET

http://localhost:8032/vuls/naive?limit=10

Send

Save

ParamsAuthorizationHeaders(9)BodyPre-request ScriptTestsSettingsCookies

Query Params

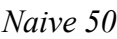
Key	Value	Bulk Edit
☒ limit	10	
Key	Value	

BodyCookies(1)Headers(5)Test Results

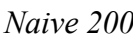
Status: 200 OKTime: 23 msSize: 1022 BSave Response

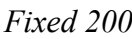
PrettyRawPreviewVisualizeJSON

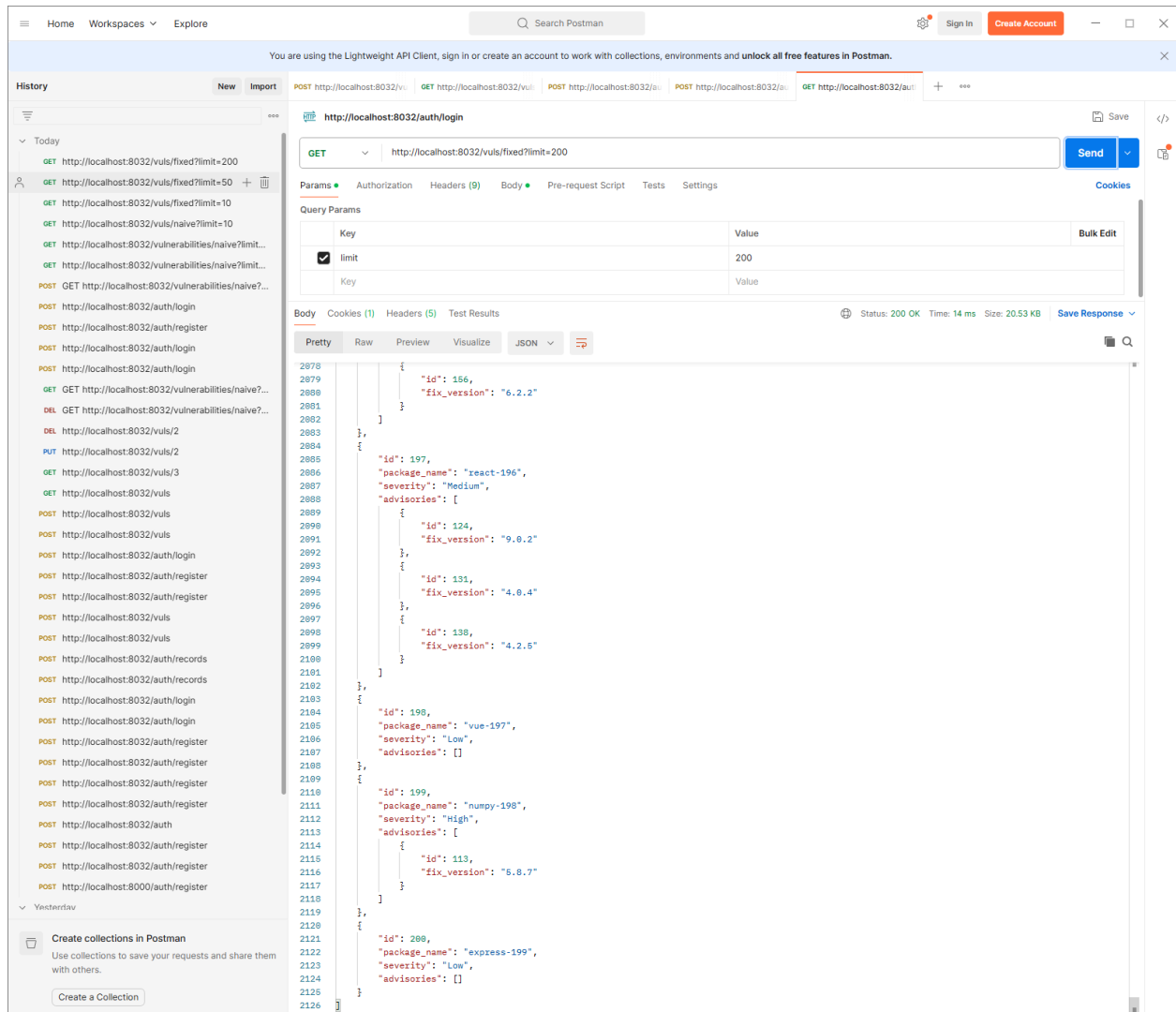
```
42      {
43        "id": 141,
44        "fix_version": "2.3.6"
45      }
46    ],
47  },
48  {
49    "id": 7,
50    "package_name": "requests-6",
51    "severity": "High",
52    "advisories": [
53      {
54        "id": 116,
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995  {
996    "id": 83,
997    "package_name": "requests-83",
1000    "severity": "High",
1001    "advisories": [
1002      {
1003        "id": 83,
1004        "fix_version": "1.3.3"
1005      }
1006    ]
1007  },
1008  {
1009    "id": 84,
1010    "package_name": "requests-84",
1011    "severity": "High",
1012    "advisories": [
1013      {
1014        "id": 84,
1015        "fix_version": "1.3.3"
1016      }
1017    ]
1018  },
1019  {
1020    "id": 85,
1021    "package_name": "requests-85",
1022    "severity": "High",
1023    "advisories": [
1024      {
1025        "id": 85,
1026        "fix_version": "1.3.3"
1027      }
1028    ]
1029  },
1030  {
1031    "id": 86,
1032    "package_name": "requests-86",
1033    "severity": "High",
1034    "advisories": [
1035      {
1036        "id": 86,
1037        "fix_version": "1.3.3"
1038      }
1039    ]
1040  },
1041  {
1042    "id": 87,
1043    "package_name": "requests-87",
1044    "severity": "High",
1045    "advisories": [
1046      {
1047        "id": 87,
1048        "fix_version": "1.3.3"
1049      }
1050    ]
1051  },
1052  {
1053    "id": 88,
1054    "package_name": "requests-88",
1055    "severity": "High",
1056    "advisories": [
1057      {
1058        "id": 88,
1059        "fix_version": "1.3.3"
1060      }
1061    ]
1062  },
1063  {
1064    "id": 89,
1065    "package_name": "requests-89",
1066    "severity": "High",
1067    "advisories": [
1068      {
1069        "id": 89,
1070        "fix_version": "1.3.3"
1071      }
1072    ]
1073  },
1074  {
1075    "id": 90,
1076    "package_name": "requests-90",
1077    "severity": "High",
1078    "advisories": [
1079      {
1080        "id": 90,
1081        "fix_version": "1.3.3"
1082      }
1083    ]
1084  },
1085  {
1086    "id": 91,
1087    "package_name": "requests-91",
1088    "severity": "High",
1089    "advisories": [
1090      {
1091        "id": 91,
1092        "fix_version": "1.3.3"
1093      }
1094    ]
1095  },
1096  {
1097    "id": 92,
1098    "package_name": "requests-92",
1099    "severity": "High",
1100    "advisories": [
1101      {
1102        "id": 92,
1103        "fix_version": "1.3.3"
1104      }
1105    ]
1106  },
1107  {
1108    "id": 93,
1109    "package_name": "requests-93",
1110    "severity": "High",
1111    "advisories": [
1112      {
1113        "id": 93,
1114        "fix_version": "1.3.3"
1115      }
1116    ]
1117  },
1118  {
1119    "id": 94,
1120    "package_name": "requests-94",
1121    "severity": "High",
1122    "advisories": [
1123      {
1124        "id": 94,
1125        "fix_version": "1.3.3"
1126      }
1127    ]
1128  },
1129  {
1130    "id": 95,
1131    "package_name": "requests-95",
1132    "severity": "High",
1133    "advisories": [
1134      {
1135        "id": 95,
1136        "fix_version": "1.3.3"
1137      }
1138    ]
1139  },
1140  {
1141    "id": 96,
1142    "package_name": "requests-96",
1143    "severity": "High",
1144    "advisories": [
1145      {
1146        "id": 96,
1147        "fix_version": "1.3.3"
1148      }
1149    ]
1150  },
1151  {
1152    "id": 97,
1153    "package_name": "requests-97",
1154    "severity": "High",
1155    "advisories": [
1156      {
1157        "id": 97,
1158        "fix_version": "1.3.3"
1159      }
1160    ]
1161  },
1162  {
1163    "id": 98,
1164    "package_name": "requests-98",
1165    "severity": "High",
1166    "advisories": [
1167      {
1168        "id": 98,
1169        "fix_version": "1.3.3"
1170      }
1171    ]
1172  },
1173  {
1174    "id": 99,
1175    "package_name": "requests-99",
1176    "severity": "High",
1177    "advisories": [
1178      {
1179        "id": 99,
1180        "fix_version": "1.3.3"
1181      }
1182    ]
1183  },
1184  {
1185    "id": 100,
1186    "package_name": "requests-100",
1187    "severity": "High",
1188    "advisories": [
1189      {
1190        "id": 100,
1191        "fix_version": "1.3.3"
1192      }
1193    ]
1194  },
1195  {
1196    "id": 101,
```











```
data260-0932 > reports > hw04 > METRICS.md
```

	Page size	Version	p50 (ms)	p95 (ms)	p99 (ms)	SQL stmts/req
1	10	naive	8.63	17.58	18.08	11
2	10	fixed	4.61	4.95	5.18	1
3	50	naive	26.15	29.56	57.69	51
4	50	fixed	5.78	6.51	6.74	1
5	200	naive	98.95	127.4	135.56	201
6	200	fixed	10.23	14.25	49.59	1

Part 4 - Build a Grounded RAG Question-Answering System

Build a small Retrieval-Augmented Generation (RAG) application that answers questions using only a supplied document collection and use it to study context engineering. Prompt engineering is how you ask the model; context engineering is what evidence you place in the model's window - the retrieved chunks, their ordering and de-duplication, and the token budget.

1. Corpus and index: use at least 5 documents (course notes, policies, or technical docs). Chunk them (start with `chunk_size = 500`, `chunk_overlap = 50`), storing each chunk's text, source name, and `chunk_id`; embed the chunks and load them into a vector store (FAISS, Chroma, or LlamaIndex VectorStoreIndex).

```
PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04> python .\rag.py
MODEL = qwen3:8b | SEED = 932 | chunks = 21 | chunk_size = 500 | overlap = 50
```

```
#read every document to cut it into chunks
chunks = []
DOCS = os.path.join(BASE, "docs")
for source in sorted(os.listdir(DOCS)):
    if not source.endswith(".txt"):
        continue
    path = os.path.join(DOCS, source)
    source = os.path.basename(path)
    text = open(path, encoding="utf-8").read()
    start = 0
    number = 0
    while True:
        chunks.append({
            "chunk_id": source + "#" + str(number),
            "source": source,
            "text": text[start:start + CHUNK_SIZE],
        })
        number = number + 1
        if start + CHUNK_SIZE >= len(text):
            break
        start = start + CHUNK_SIZE - CHUNK_OVERLAP

#chunk into a vector
vectorizer = TfidfVectorizer(stop_words="english")
vectors = vectorizer.fit_transform([c["text"] for c in chunks]).toarray().astype("float32")
faiss.normalize_L2(vectors)
index = faiss.IndexFlatIP(vectors.shape[1])
index.add(vectors)

#find the top-k chunks for a question, with their scores
def retrieve(question, k):
    q = vectorizer.transform([question]).toarray().astype("float32")
    faiss.normalize_L2(q)
    scores, ids = index.search(q, k)
    results = []
    for score, i in zip(scores[0], ids[0]):
        results.append({
            "chunk_id": chunks[i]["chunk_id"],
            "source": chunks[i]["source"],
            "text": chunks[i]["text"],
            "score": round(float(score), 3),
        })
    return results
```

2. Retrieval: for each question, retrieve the top-k chunks (start with `top_k = 3`) and print the retrieved chunks with source and score before calling the LLM, so a retrieval problem is visible separately from a generation problem.

```
#find the top-k chunks for a question, with their scores
def retrieve(question, k):
    q = vectorizer.transform([question]).toarray().astype("float32")
    faiss.normalize_L2(q)
    scores, ids = index.search(q, k)
    results = []
    for score, i in zip(scores[0], ids[0]):
        results.append({
            "chunk_id": chunks[i]["chunk_id"],
            "source": chunks[i]["source"],
            "text": chunks[i]["text"],
            "score": round(float(score), 3),
        })
    return results

def print_retrieval(qid, question, k, results):
    show(" == " + qid + " (top_k = " + str(k) + "): " + question)
    for n, r in enumerate(results, 1):
        show("#" + str(n) + " score=" + str(r["score"]) + " source=" + r["source"] + " chunk_id=" + r["chunk_id"])
        show("      " + r["text"].replace("\n", " ")[:150] + " ... ")
```

3. Run the same questions through three configurations:

- o (A) No RAG - question straight to the LLM, as a baseline.
- o (B) Basic RAG - top-3 raw chunks in the prompt.
- o (C) Context-engineered RAG - drop irrelevant and duplicate chunks, order and label the survivors with their sources, and add grounding rules: answer only from the context, cite the source number, and refuse with "I cannot answer this question from the provided documents" when the evidence is insufficient.

```
#drop chunks with a low score
def word_overlap(a, b):
    words_a = set(a.lower().split())
    words_b = set(b.lower().split())
    return len(words_a & words_b) / len(words_a | words_b)

def clean_chunks(results):
    kept = []
    for r in results:
        if r["score"] < MIN_SCORE:
            continue
        duplicate = False
        for k in kept:
            if word_overlap(r["text"], k["text"]) > 0.4:
                duplicate = True
        if not duplicate:
            kept.append(r)
    return kept

#A = no RAG, B = basic RAG (raw top-k chunks), C = context-engineered RAG (cleaned chunks, labelled with their source,
def prompt_a(question):
    return question

def prompt_b(question, results):
    context = "\n\n".join(r["text"] for r in results)
    return "Context:\n" + context + "\n\nQuestion: " + question + "\n\nAnswer:"

def prompt_c(question, results):
    context = ""
    for n, r in enumerate(results, 1):
        context = context + "[" + str(n) + "] (source: " + r["source"] + ")\n" + r["text"] + "\n\n"
    if context == "":
        context = "(no relevant context found) \n\n"
    rules = ("Rules:\n"
            "1. Answer only from the context below.\n"
            "2. After each fact, cite the source number like [1]. \n"
            "3. If the context does not have enough evidence, reply exactly: " + REFUSAL + "\n\n")
    return rules + "Context:\n" + context + "Question: " + question + "\n\nAnswer:"
```

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04> python .\rag.py
MODEL = qwen3:8b | SEED = 932 | chunks = 21 | chunk size = 500 | overlap = 50
=== Q1 (top_k = 3): How many days do we have to fix a Medium severity vulnerability?
#1 score=0.375 source=dependencies.txt chunk_id=dependencies.txt#0
    Dependency Update Policy Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
#2 score=0.333 source=severity_scoring.txt chunk_id=severity_scoring.txt#0
    Severity Scoring Guide Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
#3 score=0.272 source=dependencies.txt chunk_id=dependencies.txt#1
    severity vulnerabilities must be fixed within 14 days. Medium severity vulnerabilities must be fixed within 30 days. Low severity vulnerabilities are fixed ...
correct retrieval: yes | relevant chunks: 1 | irrelevant chunks: 2
=== Q2 (top_k = 1): What CVSS score makes a vulnerability Critical, and how many days do we have to fix it?
#1 score=0.469 source=severity_scoring.txt chunk_id=severity_scoring.txt#0
    Severity Scoring Guide Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
#2 score=0.428 source=severity_scoring.txt chunk_id=severity_scoring.txt#1
    0.1 to 3.9 is Low. A score from 4.0 to 6.9 is Medium. A score from 7.0 to 8. is High. A score from 9.0 to 10.0 is Critical. Reading the score ... The base ...
#3 score=0.29 source=dependencies.txt chunk_id=dependencies.txt#0
    Dependency Update Policy Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
correct retrieval: yes | relevant chunks: 2 | irrelevant chunks: 1
=== Q2 (top_k = 3): What CVSS score makes a vulnerability Critical, and how many days do we have to fix it?
#1 score=0.469 source=severity_scoring.txt chunk_id=severity_scoring.txt#0
    Severity Scoring Guide Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
#2 score=0.428 source=severity_scoring.txt chunk_id=severity_scoring.txt#1
    0.1 to 3.9 is Low. A score from 4.0 to 6.9 is Medium. A score from 7.0 to 8. is High. A score from 9.0 to 10.0 is Critical. Reading the score ... The base ...
#3 score=0.29 source=dependencies.txt chunk_id=dependencies.txt#0
    Dependency Update Policy Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
#4 score=0.23 source=severity_scoring.txt chunk_id=severity_scoring.txt#2
    tabase to find the CVSS score of a vulnerability. Examples. A remote code execution bug that needs no login usually scores 9.8 and is Critical. A critical ...
#5 score=0.144 source=vul_disclosure.txt chunk_id=vul_disclosure.txt#1
    r if we accept it. Coordinated disclosure. We ask reporters to keep the problem private for 90 days while we prepare a fix. If the fix is ready early ...
correct retrieval: yes | relevant chunks: 2 | irrelevant chunks: 3
=== Q3 (top_k = 3): What is the deadline for fixing a Critical vulnerability?
#1 score=0.218 source=incidents.txt chunk_id=incidents.txt#0
    Incident Response Playbook Purpose. This playbook is used when a vulnerability in an open source package is being attacked in the real world, also called ...
#2 score=0.192 source=severity_scoring.txt chunk_id=severity_scoring.txt#0
    Severity Scoring Guide Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
#3 score=0.168 source=dependencies.txt chunk_id=dependencies.txt#0
    Dependency Update Policy Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
correct retrieval: no | relevant chunks: 1 | irrelevant chunks: 2
=== Q4 (top_k = 3): How long do we have to respond?
#1 score=0.146 source=packages_track.txt chunk_id=packages_track.txt#2
    rom the lockfile so they do not create extra risk. Build systems. Build servers must run in a clean environment and must not reuse caches from other ...
#2 score=0.0 source=dependencies.txt chunk_id=dependencies.txt#2
    approve a written exception. Every exception must be reviewed again after 30 days. Testing updates. Every dependency update must pass the full automa ...
#3 score=0.0 source=dependencies.txt chunk_id=dependencies.txt#1
    severity vulnerabilities must be fixed within 14 days. Medium severity vulnerabilities must be fixed within 30 days. Low severity vulnerabilities are fixed ...
correct retrieval: no | relevant chunks: 0 | irrelevant chunks: 3
=== Q5 (top_k = 3): What is the yearly budget of the security team?
#1 score=0.274 source=vul_disclosure.txt chunk_id=vul_disclosure.txt#0
    Vulnerability Disclosure Policy Purpose. This policy explains how people outside our team can report a security problem in the open source project and which ...
correct retrieval: n/a | relevant chunks: 0 | irrelevant chunks: 3
=== Q6 (top_k = 3): How do I bake a chocolate cake?
#1 score=0.0 source=dependencies.txt chunk_id=dependencies.txt#2
    approve a written exception. Every exception must be reviewed again after 30 days. Testing updates. Every dependency update must pass the full automa ...
#2 score=0.0 source=dependencies.txt chunk_id=dependencies.txt#1
    severity vulnerabilities must be fixed within 14 days. Medium severity vulnerabilities must be fixed within 30 days. Low severity vulnerabilities are fixed ...
#3 score=0.0 source=dependencies.txt chunk_id=dependencies.txt#0
    Dependency Update Policy Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
correct retrieval: n/a | relevant chunks: 0 | irrelevant chunks: 3

```

4. Test set of six questions:

- o Q1: answer in one chunk
- o Q2: answer needs two chunks
- o Q3: similar information across documents
- o Q4: ambiguous

- o Q5: answer not in the documents
- o Q6: unrelated
- Q5 and Q6 must be refused, not answered.

```
questions = [
    {"id": "Q1", "type": "answer in one chunk", "question": "How many days do we have to fix a Medium severity vulnerability?", "key_facts": ["30 days"], "answer_all": ["30"], "answer_any": [], "must_refuse": false},
    {"id": "Q2", "type": "answer needs two chunks", "question": "What CVSS score makes a vulnerability Critical, and how many days do we have to fix it?", "key_facts": ["9.0 to 10.0", "within 7 days"], "answer_all": ["9.0", "7"], "answer_any": [], "must_refuse": false},
    {"id": "Q3", "type": "similar information across documents", "question": "What is the deadline for fixing a critical vulnerability?", "key_facts": ["within 7 days", "within 24 hours"], "answer_all": ["7 days", "24 hours"], "answer_any": [], "must_refuse": false},
    {"id": "Q4", "type": "ambiguous", "question": "How long do we have to respond?", "key_facts": ["3 business days"], "answer_all": [], "answer_any": ["which", "clarify", "specify", "depends", "different", "several"], "must_refuse": false},
    {"id": "Q5", "type": "answer not in the documents", "question": "What is the yearly budget of the security team?", "key_facts": [], "answer_all": [], "answer_any": [], "must_refuse": true},
    {"id": "Q6", "type": "unrelated", "question": "How do I bake a chocolate cake?", "key_facts": [], "answer_all": [], "answer_any": [], "must_refuse": true},
]
```

```
Q1 A -> The number of days required to fix a Medium severity vulnerability depends on organizational pol
Q1 B -> The policy states that Medium severity vulnerabilities must be fixed within 30 days. Answer:
Q1 C -> The team must fix a Medium severity vulnerability within 30 days [1].
Q2 A -> The CVSS (Common Vulnerability Scoring System) defines a vulnerability as Critical if its CVSS
Q2 B -> A vulnerability is classified as Critical if its CVSS score falls within the range 9.0 to 10.0
Q2 C -> A CVSS score of 9.0 to 10.0 makes a vulnerability Critical [2]. The context does not specify how man
Q3 A -> The deadline for fixing a Critical vulnerability depends on several factors, including industry
Q3 B -> The deadline for fixing a Critical vulnerability is within 7 days. This is specified in the De
Q3 C -> I cannot answer this question from the provided documents. The context mentions an "Emergency deadli
Q4 A -> The time frame for responding depends on the specific context. Here are some common scenarios and ty
Q4 B -> The response time depends on the severity of the vulnerability: - High severity vulnerabilities
Q4 C -> I cannot answer this question from the provided documents.
Q5 A -> The yearly budget of a security team can vary significantly depending on the organization's size, in
Q5 B -> The provided context does not mention or specify the yearly budget of the security team. The documen
Q5 C -> I cannot answer this question from the provided documents.
Q6 A -> Baking a chocolate cake is a rewarding process! Here's a step-by-step guide to help you create a moi
Q6 B -> The context provided is about a software dependency update policy and does not relate to baking a ch
Q6 C -> I cannot answer this question from the provided documents.
```

- 5. Sweep the context size: run at least one question at $k = 1, 3$, and 5 , and report whether more context helped, when irrelevant chunks appeared, and which k was best**
- more context is not always better context.

```
q = questions[1]
sweep_rows = []
for k in [1, 3, 5]:
    raw_results = retrieve(q["question"], k)
    clean_results = clean_chunks(raw_results)
    for config in ["B", "C"]:
        if config == "B":
            used, prompt = raw_results, prompt_b(q["question"], raw_results)
        else:
            used, prompt = clean_results, prompt_c(q["question"], clean_results)
        answer = ask_llm(prompt)
        relevant = [r for r in used if facts_found([r], q["key_facts"])]
        print(f"sweep k = {k}, config, {config}, {len(relevant)}\n")
        sweep_rows.append([k, config, len(used), len(relevant), len(used) - len(relevant), is_correct(q, answer), answer.replace(

with open(RAW + "/rag_k_sweep.csv", "w", newline="", encoding="utf-8") as f:
    w = csv.writer(f)
    w.writerow(["top_k", "config", "chunks_in_prompt", "relevant", "irrelevant", "correct_answer", "answer"])
    w.writerows(sweep_rows)

summary = []
for config in ["A", "B", "C"]:
    rows = [r for r in eval_rows if r[1] == config]
    accuracy = sum(1 for r in rows if r[3]) / 6
    faithful = sum(1 for r in rows if r[4]) / 6
    robust = sum(1 for r in rows if r[0] in ("Q5", "Q6") and r[5] is True) / 2
    fmt = sum(1 for r in rows if r[6] is True) / 6 if config == "C" else "n/a"
    summary.append([config, round(accuracy, 2), round(faithful, 2), fmt if fmt == "n/a" else round(fmt, 2), robust])
```

```
sweep k = 1 B -> Answer: A vulnerability is classified as Critical if its CVSS score is 9.0 or higher (
sweep k = 1 C -> I cannot answer this question from the provided documents. The context does not specify the CVSS sco
sweep k = 3 B -> A vulnerability is classified as Critical if its CVSS score falls within the range 9.0 to 10.0
sweep k = 3 C -> A CVSS score of 9.0 to 10.0 makes a vulnerability Critical [2]. The context does not specify how man
sweep k = 5 B -> A vulnerability is classified as Critical if its CVSS score falls within the range 9.0 to 10.0
sweep k = 5 C -> A CVSS score of 9.0 to 10.0 makes a vulnerability Critical [1][2]. The context mentions a 90-day per
```

- 6. Evaluate:** for each question and configuration, record correct retrieval, correct answer, grounded (answer supported by the cited chunks), and refused-when-needed; summarize accuracy, faithfulness, format compliance, and robustness in one table.

```

summary = []
for config in ["A", "B", "C"]:
    rows = [r for r in eval_rows if r[1] == config]
    accuracy = sum(1 for r in rows if r[3]) / 6
    faithful = sum(1 for r in rows if r[4]) / 6
    robust = sum(1 for r in rows if r[0] in ("Q5", "Q6") and r[5] is True) / 2
    fmt = sum(1 for r in rows if r[6] is True) / 6 if config == "C" else "n/a"
    summary.append([config, round(accuracy, 2), round(faithful, 2), fmt if fmt == "n/a" else round(fmt, 2), robust])

with open(RAW + "/rag_evaluation_summary.csv", "w", newline="", encoding="utf-8") as f:
    w = csv.writer(f)
    w.writerow(["config", "accuracy", "faithfulness", "format_compliance", "robustness"])
    w.writerows(summary)
print("config | accuracy | faithfulness | format_compliance | robustness")
for row in summary:
    print(row)

```

```

config | accuracy | faithfulness | format_compliance | robustness
['A', 0.5, 0.0, 'n/a', 0.0]
['B', 0.5, 0.5, 'n/a', 0.0]
['C', 0.5, 0.5, 1.0, 1.0]

```

7. Write a short analysis (300–500 words): which questions retrieved the right chunks, which context changes helped (top_k, de-duplication, chunk size, source metadata, ordering, shorter context, grounding rules), whether the model hallucinated on Q5/Q6, and how retrieval quality, context quality, and the prompt together shaped answer quality.

Question 1 got the right chunks defining the 30 day rule as the answer from dependencies.txt file. Q2 retrieved two chunks and got the severity level chunk. It only missed the score ranges for the severity levels. Q3 was retrieving the same information for all the files where the response time should be within 24hrs. Q4 there were issues because it treated respond and response with different words and not the say. Therefore, the policy was not retrieved correctly. Q5 did not work because there was no answer in the documents as well as Q6. Q6 was a question that was unrelated so it refused to answer too. Some context changes that helped was the grounding rules the most. I found that C was able to get the required sentence that it could not answer. I think for top_k, the value of 3 worked best for Q2. The lower value of 1 did not get enough chunks, while the higher value of 5 got irrelevant chunks. I had two documents called dependencies.txt and severity_scoring.txt that had duplicate paragraphs talking about security levels. With de duplication, I found that it negatively impacted the models because it could not find the right information. I think it was because it checked the wrong place and that is why some of the results showed it couldn't answer it correctly. For hallucination, I found that Config A gave a very different answer instead of saying it could not find the information. The best config was C because it always answered it could not find the information. As for retrieval quality, I found that prompts could not fix the error of the chunk with the correct answer missing. However, I did find that basic RAG was able to answer correctly when the retrieval is good. From this, I can conclude that the most important thing is the better the chunking and then the prompt. All three configurations had about the same context quality. However, the main point is that the chunking is what affected all the configurations and how they performed. Prompting can also improve the output when you ask it more simply. This gives it less room for assumptions and overcomplications.

Save your retrieved-chunk printouts

```
data260-0932 > reports > hw04 > raw > rag_retrieval_printouts.txt
1  MODEL = qwen3:8b | SEED = 932 | chunks = 21 | chunk_size = 500 | overlap = 50
2  === Q1 (top_k = 3): How many days do we have to fix a Medium severity vulnerability?
3  #1 score=0.375 source=dependencies.txt chunk_id=dependencies.txt#0
4    Dependency Update Policy Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
5  #2 score=0.333 source=severity_scoring.txt chunk_id=severity_scoring.txt#0
6    Severity Scoring Guide Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS s ...
7  #3 score=0.272 source=dependencies.txt chunk_id=dependencies.txt#1
8    ertainty vulnerabilities must be fixed within 14 days. Medium severity vulnerabilities must be fixed within 30 days. Low severity vulnerabilities are f ...
9  correct retrieval: yes | relevant chunks: 1 | irrelevant chunks: 2
10 === Q2 (top_k = 1): What CVSS score makes a vulnerability Critical, and how many days do we have to fix it?
11 #1 score=0.469 source=severity_scoring.txt chunk_id=severity_scoring.txt#0
12    Severity Scoring Guide Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS s ...
13 correct retrieval: no | relevant chunks: 0 | irrelevant chunks: 1
14 === Q2 (top_k = 3): What CVSS score makes a vulnerability Critical, and how many days do we have to fix it?
15 #1 score=0.469 source=severity_scoring.txt chunk_id=severity_scoring.txt#0
16    Severity Scoring Guide Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS s ...
17 #2 score=0.428 source=severity_scoring.txt chunk_id=severity_scoring.txt#1
18    0.1 to 3.9 is Low. A score from 4.0 to 6.9 is Medium. A score from 7.0 to 8. is High. A score from 9.0 to 10.0 is Critical. Reading the score. The ba ...
19 #3 score=0.29 source=dependencies.txt chunk_id=dependencies.txt#0
20    Dependency Update Policy Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS ...
21 correct retrieval: yes | relevant chunks: 2 | irrelevant chunks: 1
22 === Q2 (top_k = 5): What CVSS score makes a vulnerability Critical, and how many days do we have to fix it?
23 #1 score=0.469 source=severity_scoring.txt chunk_id=severity_scoring.txt#0
24    Severity Scoring Guide Severity levels. Every vulnerability is given a severity level: Low, Medium, High or Critical. The level comes from the CVSS s ...
```

the three-configuration comparison

```
data260-0932 > reports > hw04 > raw > {} rag_comparison.json > ...
1  {
2    "model": "qwen3:8b",
3    "seed": 932,
4    "results": [
5      {
6        "question": "Q1",
7        "config": "A",
8        "chunks_used": [],
9        "answer": "The number of days required to fix a **Medium severity vulnerability** depends on organizational policies, industry standards, and the specific context of
10      },
11      {
12        "question": "Q1",
13        "config": "B",
14        "chunks_used": [
15          "dependencies.txt#0",
16          "severity_scoring.txt#0",
17          "dependencies.txt#1"
18        ],
19        "answer": "The policy states that **Medium severity vulnerabilities must be fixed within 30 days**.\n\n**Answer:** 30 days."
20      },
21      {
22        "question": "Q1",
23        "config": "C",
24        "chunks_used": [
25          "dependencies.txt#0",
26          "dependencies.txt#1"
27        ],
28        "answer": "The team must fix a Medium severity vulnerability within 30 days [1]."
29      },
30      {
31        "question": "Q2",
32        "config": "A",
33        "chunks_used": [],
34        "answer": "The **CVSS (Common Vulnerability Scoring System)** defines a vulnerability as **Critical** if its **CVSS score is 9.0 or higher** (in CVSS v3.1, which is the current sta
35      },
36      {
37        "question": "Q2",
38        "config": "B",
39        "chunks_used": [
40          "severity_scoring.txt#0",
41          "severity_scoring.txt#1",
42          "dependencies.txt#0"
43        ],
44        "answer": "A vulnerability is classified as **Critical** if its CVSS score falls within the range **9.0 to 10.0**. Critical vulnerabilities must be fixed within **7 days** [1].
45      },
46      {
47        "question": "Q2",
48        "config": "C",
49        "chunks_used": [
50          "severity_scoring.txt#0",
51          "severity_scoring.txt#1"
52        ],
53        "answer": "A CVSS score of 9.0 to 10.0 makes a vulnerability Critical [2]. The context does not specify how many days are allowed to fix it [1].\n\nI cannot answer this question f
54      },
55      {
56        "question": "Q3",
57        "config": "A",
58        "chunks_used": [],
```

the k-sweep results

```

1 top_k,config,chunks_in_prompt,relevant,irrelevant,correct_answer,answer
2 1,8,1,0,1,False,"**Answer:** A vulnerability is classified as **Critical** if its CVSS score is **7.0 or higher** (ranging from 7.0 to 10.0). The exact number of days to
3 1,C,1,0,1,False,"I cannot answer this question from the provided documents. The context does not specify the CVSS score range for ""Critical"" or the timeframe for fixing v
4 3,8,3,2,1,True,"A vulnerability is classified as **Critical** if its CVSS score falls within the range **9.0 to 10.0**". According to the Dependency Update Policy, **Critici
5 3,C,2,1,1,False,"A CVSS score of 9.0 to 10.0 makes a vulnerability Critical [2]. The context does not specify how many days are allowed to fix a Critical vulnerability [1].
6 5,8,5,2,3,True,"A vulnerability is classified as **Critical** if its CVSS score falls within the range **9.0 to 10.0**". Critical vulnerabilities must be fixed within **7 day
7 5,C,4,1,3,False,"A CVSS score of 9.0 to 10.0 makes a vulnerability Critical [1][2]. The context mentions a 90-day period for coordinated disclosure while preparing a fix, l
8

```

and the evaluation table in reports/hw04/raw/, and include your written analysis in report.pdf.

```

1 config,accuracy,faithfulness,format_compliance,robustness
2 A,0.5,0.0,n/a,0.0
3 B,0.5,0.5,n/a,0.0
4 C,0.5,0.5,1.0,1.0
5

```

```

1 question,config,correct_retrieval,correct_answer,grounded,refused_when_needed,format_ok
2 Q1,A,n/a,True,False,n/a,n/a
3 Q1,B,yes,True,True,n/a,n/a
4 Q1,C,yes,True,True,n/a,True
5 Q2,A,n/a,True,False,n/a,n/a
6 Q2,B,yes,True,True,n/a,n/a
7 Q2,C,no,False,False,n/a,True
8 Q3,A,n/a,False,False,n/a,n/a
9 Q3,B,no,False,False,n/a,n/a
10 Q3,C,no,False,False,n/a,True
11 Q4,A,n/a,True,False,n/a,n/a
12 Q4,B,no,True,True,n/a,n/a
13 Q4,C,no,False,False,n/a,True
14 Q5,A,n/a,False,False,False,n/a
15 Q5,B,n/a,False,False,False,n/a
16 Q5,C,n/a,True,True,True,True
17 Q6,A,n/a,False,False,False,n/a
18 Q6,B,n/a,False,False,False,n/a
19 Q6,C,n/a,True,True,True,True
20

```

Verification output

```

PS C:\Users\natal\Documents\masters\fall 26\data 260\github\data260-0932\reports\hw04> python .\verify_hw04.py
PASS backend responds on PORT_BASE -GET /health returned 200
PASS list endpoint needs login -GET /vuls without login returned 401
PASS login sets an HTTP-only session cookie -login returned 200, HTTP-only session cookie set: True
PASS naive list endpoint returns data -limit 50 -> 50 records
PASS fixed list endpoint returns data -limit 50 -> 50 records
PASS naive and fixed return the same data -naive and fixed return the same records: True
PASS raw/ has all 180 N+1 requests -raw/n_plus_one_raw.csv has 180 rows
PASS RAG corpus has at least 5 documents -6 documents in rag/docs

```

AI_USE.md

```

data260-0932 > reports > hw04 > AI_USE.md
1 1. I used the AI assistant to help me with debugging rag.py file. I also used it to help debug the verify_hw04.py file to
2 check the connection via port and seed because I was unfamiliar with checking react. I created the base rag.py file that
3 does the chunking and the three different approaches. I had issues with rag not displaying my output files because it was
4 not in the right folder directory. I accidentally set it to write to the /raw folder in my C drive because I did not specify
5 the path correctly. I used it for verify python to help figure out how to make sure the commit was checked and used in the
6 code as specified in the report pdf.
7
8 2. One thing the AI produced was that it told me to that the removing duplicates gave an accuracy error that was way lower
9 than it should be. I found that the cosine similarity after removing duplicates was odd so I looked into it and fixed that
10 step. The code still gives some chunking issues that I have fixed. So the outputs was able to give me the specific outputs
11 each of the six questions ask for. It also reports correctly when the it doesn't have any information.
12
13 3. I detected this problem because C said on question 2 the context did not say the correct day while B did. This led me to
14 realize the chunking was off. I verified it by showing the before chunking and after chunking after duplication to figure
15 out what the issue was.
16
17 4. I changed it so that the threshold for removing duplicates so that the duplicate chunks are removed without removing
18 anything extra. I think the threshold was causing the error. So I changed it to above 0.4 and that gave me a better
19 result. I verified this when I reran the command python rag.py. |

```